

Product Relatedness in Tasks: A Novel Measure and Multi Product Firms

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Abstract

Products vary in the tasks they require and, consequently, in the occupations they draw on. I develop a novel measure of product relatedness grounded in the occupational composition of single product manufacturers in Denmark. Using matched employer–employee data, I represent each product as a vector of task requirements and construct a similarity measure that combines occupational overlap with skill transferability inferred from worker flows. To take the measure to multi product firms, I anchor each firm to its core product and compute the task distance between that core and every candidate product, linking these distances to the firm’s product-scope decisions. Multi product firms expand into products closely related to their core and drop products that are more distant in task space, consistent with rising marginal costs of producing away from core competencies. These findings highlight the role of workforce composition in shaping firm scope and offer a task-based perspective on product diversification.

Keywords: Multi Product Firms, Product Choice, Core Competences, Task Similarity, Product Relatedness, Occupations

JEL codes: L23, L25, J24

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1 Introduction

Multi product firms (MPFs) account for a disproportionately large share of production, exports, and employment¹, yet we still know relatively little about how they make decisions on *which* products to add (drop) in the product mix from an empirical perspective. This paper studies product diversification (retraction) from the perspective of a firm’s labor-based core competences. The central idea is simple: products differ in the occupations and tasks required to produce them, and firms should find it easier to expand into products that draw on capabilities already present in their workforce.

This perspective is closely related to classic resource-based views (RBV) of the firm, which place the extensive margin of production as a key driver of growth. Specifically, in Penrose (1959), firm growth reflects the accumulation and redeployment of internal resources in new product lines, while Schumpeter (1951) emphasizes innovation and product diversification through the recombination of existing capabilities coined ”creative destruction”. The RBV literature further highlights flexibility in manufacturing and dynamic capabilities as sources of competitive advantage, especially in times of increasing globalization and rapid technological change (Teece et al., 1997). Applied to MPFs, these ideas imply that product diversification is shaped not only by demand conditions, but also by whether firms can reuse and reconfigure what they already know how to do. A central theme in the MPF literature in this context is that firms actively adjust scope around a set of core products or core competences (as in Eckel and Neary (2010) for instance). The question, then, is how to measure those underlying capabilities in a way that is observable in the data.

In this paper, I propose to do so using occupational structures. Occupations provide a tractable empirical proxy for the tasks required for production. If two products rely on similar occupational compositions, they are likely to require similar tasks and capabilities; if their occupational structures differ substantially, moving between them should be more

¹Bernard et al. (2005) shows that product scope and choice are far from random, Bernard et al. (2007) summarize the firm-level trade facts in relation to product scope, Bernard et al. (2010, 2011) provide the benchmark empirical and theoretical treatments of product switching and endogenous product scope, and subsequent work by Eckel and Neary (2010), Nocke and Yeaple (2008, 2014), and Arkolakis et al. (2021) studies how competition, globalization, and core competences shape firms’ product range and mix. See ? for a recent survey.

costly. Accordingly, I argue that product-adding decisions depend positively, and product-dropping decisions depend negatively, on occupational similarity.

The idea of linking occupations and products is not new², and the particular approach of this paper builds on several strands of related work. Farjoun (1994) show that diversification is systematically related to similarities in human expertise across industries, using occupational profiles to identify resource-related industry groups. Neffke and Henning (2013) instead infer relatedness from worker mobility, arguing that labor flows reveal overlap in underlying skill requirements. More recently, Boehm et al. (2022) show that firms are more likely to diversify into products with similar input requirements, consistent with a resource-based view of firm scope. This paper combines these insights by constructing product-level occupational profiles and using them to measure relatedness between bilateral product pairs and between firms and products via their core products. Relative to existing work, the contribution is to measure product capabilities directly from linked employer–employee data and to allow occupations to be partially related rather than either identical or distinct.

A key empirical challenge is to identify the task requirements of individual products. I address this using single product firms (SPFs), defined as firms whose activity is concentrated in one core product, while allowing for limited co-production in the primary analysis. I show that many firms produce only small amounts outside of their core product, and that these particular SPFs’ core products expand the range of SPF covered products; introducing a tradeoff between coverage and cleanness in the SPF definition. The occupational composition of these firms provides an empirical representation of the labor inputs needed to produce each product. Aggregating hours worked by occupation across SPFs within a product yields a product-level occupational vector, where each occupation’s share reflects its importance in production.

These product-level vectors are then used to construct bilateral product relatedness measures. The baseline measure is a cosine similarity index in the spirit of Boehm et al. (2022);

²There is extensive work linking workers and occupations to industries and products in relation to technological change of different kinds (Acemoglu and Autor, 2011), globalization (Autor et al., 2013; Ebenstein et al., 2014; Traiberman, 2019; Akerman and Holzheu, 2024), offshoring in particular (Becker et al., 2013; Crinò, 2010), and other shocks; for instance the Covid shock (del Rio-Chanona et al., 2020; Chopra et al., 2022; Mumtaz and Smith, 2023), health exposure (Huss et al., 2013), among many other multi-faceted analyses.

products are more related when their occupational structures are more similar. I extend this framework by allowing occupational inputs to interact through a weighting matrix based on occupation-to-occupation skill sharing. These weights are motivated by the labor-flow logic of Neffke and Henning (2013); occupations that exchange workers more easily are treated as closer in task space (see also Joyez and Laffineur (2022)). This extension is important because production processes may rely on occupations that are not identical across products, but nevertheless involve overlapping tasks.

The resulting similarity structure reveals clear and economically intuitive patterns across the product space. Products tend to cluster into distinct groups with high internal similarity, such as food-related products, plastic and rubber goods, and machinery and equipment, reflecting shared occupational requirements within these domains. At the same time, the measure uncovers strong cross-category linkages that are not apparent from standard product classifications; some products exhibit high similarity despite being distant in the product taxonomy, while others that are closely related in classification differ substantially in their occupational composition. These patterns indicate that the proposed measure captures meaningful variation in underlying task requirements rather than simply reproducing classification-based proximity.

After constructing product-to-product relatedness from SPFs, I turn to MPFs, which are excluded from the estimation of product-level occupational profiles. For each MPF, I proxy the core competence of the firm with the core product (in which the majority of sales is concentrated). This yields a firm-to-product relatedness measure that captures the relatedness of a candidate product to each firm's core capabilities.

Empirically, the organization of MPF production yields support for the idea of flexible manufacturing: the mean task similarity to the core-product of the firm decreases in product rank. Furthermore, I find clear evidence for firms do *not* adjust their product mix randomly; rather, they expand along paths that economize on the capabilities already embedded in the firm. Candidate products that are more related to a firm's existing activities are more likely to be added to its portfolio. A positive and significant linear estimate for adding probability masks a pronounced non-linear effect: product adding occurs primarily among the *very* most similar products to the core of the firm, with a sharp spike in adding probability for

the absolute top percentile in similarity. Product dropping occurs predominantly for fringe incumbent products, characterized by low sales shares and lower similarities to core products.

The paper contributes to the literature in three ways. First, it provides a new measure of product relatedness grounded in a combination of firm-product specific data and matched employer-employee information.³ Second, it offers a transparent way to recover product-level task requirements from observed data using SPFs. Third, it shows that the organization of production across multiple product lines, and product scope adjustment decisions in particular, is closely tied to workforce-based relatedness, highlighting labor composition as a central determinant of firm scope.

More broadly, the paper offers a task-based view of product diversification decisions. By linking products to the occupations required to produce them, it furthermore provides a new way to understand why some product combinations are common within firms while others are rare.

The remainder of this paper is organized as follows: Section 2 provides a simple extension to the theoretical framework of Eckel and Neary (2010) by introducing a functional form for the technological component of firm-specific marginal costs of producing candidate products. Section 3 introduces the data and provides summary statistics on firms and products, as well as workers and occupations. It also contains a section with stylized facts on the link between occupations and (adjustments of) the extensive margin of product scope. Section 4 starts by defining SPFs and continues by describing the construction of product-level occupational vectors based on these firms. Section 5 introduces the weighted cosine similarity formula for constructing bilateral product similarities and the associated weighting matrix based on bilateral occupation excess labor flows. It concludes by providing an overview of the structure of bilateral product similarities across the product space. Section 6 contains the MPF-level analysis and ties the results to theoretical predictions from Section 2. Section 7 concludes.

³Boehm et al. (2022) use both input and output firm-product data, while Neffke and Henning (2013) and Farjoun (1994) use industry-level occupational data. Other work on diversification decisions uses firm-product level data on the output market (Macedoni et al., 2024).

2 Scope Adjustments and Task Similarity in Theory

This section provides a brief extension of the MPF framework in Eckel and Neary (2010) that links firms' product scope decisions to occupational similarity across products. The goal is to derive empirically testable predictions for product adding and dropping decisions using observable measures of similarity between products. In particular, I show how occupational similarity can be interpreted as a determinant of marginal production costs, and how this feeds into firms' endogenous scope choices.

Eckel and Neary (2010) model the marginal cost of producing any product as wages (prices) multiplied by a technological component (new task requirements), which increases in the distance to a firm's core competences. Inspired by this, I let ω denote a candidate product for firm j , whose core competences are proxied by its core product $\bar{\omega}_j$.

The latent firm-level cost of adopting a product ω depends on the set of new tasks required to produce ω , which is captured by $1 - \Phi(\bar{\omega}_j, \omega)$, where $\Phi(\bar{\omega}_j, \omega) \in [0, 1]$ denotes the similarity in tasks between the firm's core product $\bar{\omega}_j$ and the candidate product. The marginal cost of producing a candidate product is thus modeled as

$$c_j(\omega) = W \cdot [\gamma_\omega^0 + \kappa (1 - \Phi(\bar{\omega}_j, \omega))^\eta]. \quad (1)$$

Here, W denotes wages, γ_ω^0 captures the baseline task requirement of ω (and may be zero), κ governs the steepness of the marginal cost function, and η determines its curvature. I maintain $\kappa > 0$ and $\eta > 0$ throughout, so that marginal cost increases in the distance to core competences; the curvature parameter η is otherwise left free and is identified in the empirical analysis. In the spirit of Eckel and Neary (2010), one can illustrate the model as displayed in Figure 2 for $\kappa > 0$ and $\eta = 1$.

Using the same aggregate demand setup as in Eckel and Neary (2010), a candidate product ω is viable if its effective (choke) price at least covers marginal cost:

$$a' - W \cdot [\gamma_\omega^0 + \kappa (1 - \Phi(\bar{\omega}_j, \omega))^\eta] - b'e(X_j + Y) \geq 0, \quad (2)$$

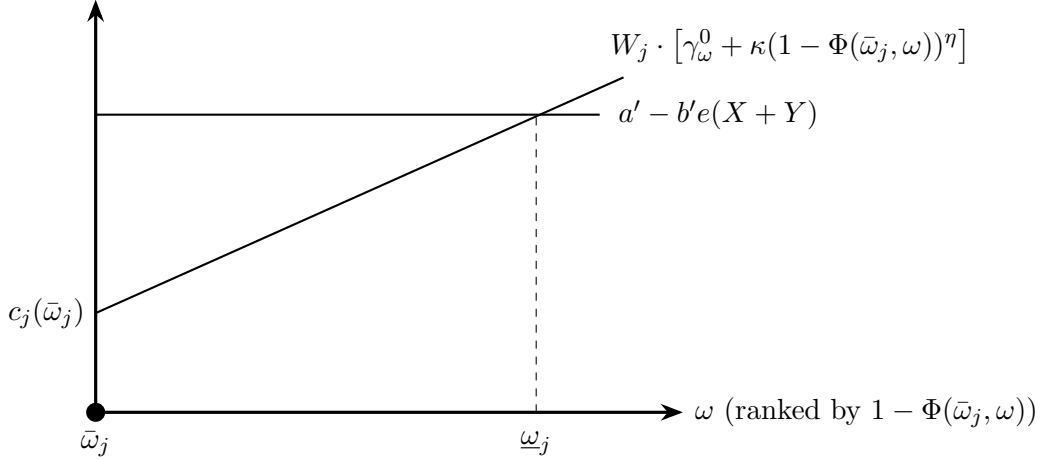


Figure 1: Authors own drawings adapted from Eckel and Neary (2010). Illustration of firm product mix decisions with flexible manufacturing: The firm's marginal costs of producing any latent product increase with distance in required tasks to the core product of the firm. A firm will produce any product where marginal costs are below marginal revenue.

where a' , b' , and e are non-negative and denote the broad willingness to pay, the slope of demand (which decreases in market size), and substitutability across products, respectively. X_j is firm j 's output of other products, capturing cannibalization effects, and Y denotes aggregate output of the product across all firms, capturing competition effects. It is convenient to collect the demand and cost shifters into the net surplus

$$s_{j\omega} \equiv a' - W\gamma_{\omega}^0 - b'e(X_j + Y), \quad (3)$$

so that Equation (2) is equivalent to the viability condition $s_{j\omega} \geq \kappa W(1 - \Phi(\bar{\omega}_j, \omega))^\eta$. The firm's product scope is the set of products satisfying $c_j(\omega) \leq a' - b'e(X_j + Y)$, which also defines the range of products over which firm j has cost advantages from flexible manufacturing, denoted $[\bar{\omega}_j, \underline{\omega}_j]$, where $\underline{\omega}_j$ is the marginal product.⁴

Rearranging Equation (2) yields a similarity threshold for production, all else equal, such that production requires $\Phi(\bar{\omega}_j, \omega) \geq \Phi_{j\omega}^*$.⁵ This threshold is the minimum similarity

⁴Under a common baseline task requirement, $\gamma_{\omega}^0 \equiv \gamma^0$, the threshold below is identical across candidates and the viable set is an interval in similarity space, $\Phi \in [\Phi_j^*, 1]$, with $\underline{\omega}_j$ attaining Φ_j^* . With product-specific γ_{ω}^0 the cutoff varies across candidates and the viable set need not be a strict interval in similarity.

⁵Setting Equation (2) to zero and solving for Φ yields the cutoff

$$\Phi_{j\omega}^* = 1 - \left(\frac{a' - b'e(X_j + Y) - W\gamma_{\omega}^0}{\kappa W} \right)^{\frac{1}{\eta}} = 1 - \left(\frac{s_{j\omega}}{\kappa W} \right)^{\frac{1}{\eta}}.$$

required for a firm to produce a candidate product. Since dropping applies only to products already in the firm's portfolio, let $\Omega_j \subseteq \mathcal{P}$ denote firm j 's current product set within the full product space $\mathcal{P}$. The adding and dropping rules are then exact mirror images, restricted to non-incumbent and incumbent products respectively:

$$\text{Add}_{j\omega} = \mathbf{1}\{\omega \notin \Omega_j\} \cdot \mathbf{1}\{\Phi(\bar{\omega}_j, \omega) \geq \Phi_{j\omega}^*\}, \quad (4)$$

$$\text{Drop}_{j\omega} = \mathbf{1}\{\omega \in \Omega_j\} \cdot \mathbf{1}\{\Phi(\bar{\omega}_j, \omega) < \Phi_{j\omega}^*\}. \quad (5)$$

The marginal product $\underline{\omega}_j$ is defined where the inequality binds. In this frictionless static benchmark a single cutoff governs both margins: a product is viable if and only if $\Phi \geq \Phi_{j\omega}^*$. The model thus abstracts from adjustment costs, and any empirical asymmetry between adding and dropping operates through differences in the joint distribution of $(\Phi, s_{j\omega})$ across the candidate and incumbent pools, rather than through separate thresholds.

To move from the deterministic rules to testable statements about probabilities, note that the demand and cost shifters entering $s_{j\omega}$ vary across firm-product pairs. I will think of the net surplus $s_{j\omega}$ as a random variable with cdf G , and equivalently the threshold $\Phi_{j\omega}^*$ is stochastic. Conditional on similarity to the core, the probability that a candidate product is added is

$$P(\text{Add}_{j\omega} = 1 \mid \Phi) = P(s_{j\omega} \geq \kappa W(1 - \Phi)^\eta) = 1 - G(\kappa W(1 - \Phi)^\eta). \quad (6)$$

Differentiating, the probability of adding increases in similarity to the firm's core product:

$$\frac{\partial P(\text{Add}_{j\omega} = 1 \mid \Phi)}{\partial \Phi} = g(\kappa W(1 - \Phi)^\eta) \kappa W \eta (1 - \Phi)^{\eta-1} > 0. \quad (7)$$

In the graphical illustration in Figure 2, this implies that products closer to $\bar{\omega}_j$ in task space are more likely to be added by firm j .

The *curvature* of this relationship is governed by η . Because distance enters through

Appendix A.1 shows that the threshold similarity is decreasing in demand and increasing in firm sales of other products, total output of the product, wages, adaptation costs, and baseline product costs. The intuition is straightforward: stronger demand lowers the effective barrier to entry, while competition, cannibalization, and cost factors raise it. The wage result relies on $s_{j\omega} > 0$, which holds for any product in the viable region.

$(1 - \Phi)^\eta$, a steep near-core cost gradient concentrates viable products among those most similar to the core, so that adding probability rises appreciably only at the very top of the similarity distribution. This nonlinearity, rather than the sign in Equation (7) alone, is the object identified by the percentile-binned specification in the empirical analysis (Section 6).⁶

Symmetrically, for incumbent products the probability of dropping is

$$P(\text{Drop}_{j\omega} = 1 \mid \Phi, \omega \in \Omega_j) = G(\kappa W(1 - \Phi)^\eta), \quad (8)$$

which decreases in similarity:

$$\frac{\partial P(\text{Drop}_{j\omega} = 1 \mid \Phi, \omega \in \Omega_j)}{\partial \Phi} < 0, \quad (9)$$

implying that incumbent products closer to the marginal product in similarity are more likely to be dropped.

⁶For example, if $s_{j\omega} \sim U[0, \bar{s}]$ over the relevant range, then $P(\text{Add}_{j\omega} = 1 \mid \Phi) = 1 - (\kappa W/\bar{s})(1 - \Phi)^\eta$ is convex in Φ ; placing the mass of additions among the most similar products, if and only if $\eta < 1$. In that case the marginal cost of the first unit of distance from the core is steep, so that only products extremely close to $\bar{\omega}_j$ are viable, producing a sharp spike in adding probability for the highest similarity percentiles.

3 Firms and Products; Workers and Occupations

This section presents the data sources, summarizes the sample, and provides novel stylized facts on co-production patterns and occupational structures in Danish manufacturing firms. It then introduces new facts about the interaction between product scope and occupational scope in Section 3.2.

3.1 Data and key variables

One methodological aim of this paper is to rely only on observed data. This requires detailed information on both firms and workers. I leverage registry data covering the universe of Danish manufacturing firms, their yearly product-level sales, and associated worker records including occupational codes. To avoid possible contamination from the financial crisis and Covid-19 mitigation policies, I restrict the sample to the period from 2010 to 2019, inclusive.

3.1.1 Firms and products

Firm-level data cover all firms registered in the Danish industry classification, Dansk Branchekode 2007 (DB07), in sectors ‘B Råstofvinding’ (Mining and quarrying) and ‘C Industri’ (Manufacturing) from the VARS database. The database contains quarterly sales data for all firms at a highly detailed level, with product codes of up to 10 digits following the Combined Nomenclature (CN). Importantly, firms are included only if they have more than 10 employees, measured in the third quarter of the year preceding the reference year. I first exclude all firms registered within Mining and quarrying, and then aggregate product-level sales to the yearly level for each firm. From the FIRM database, I add firm-year information on revenue, value added, and employment. Some firms have relatively high employment or revenue compared to the sample averages. I therefore remove outliers that report revenue or employee counts more than three standard deviations (sd) above the yearly mean in any given sample year.

The CN product codes correspond to Harmonized System (HS) product codes at the six-digit level. However, this level is too detailed for the purposes of this paper. For example,

it separates “Meat of swine” into two categories: “fresh or chilled” (020319) and “frozen” (020329). I regard this distinction as too specific for the present analysis. In the main sample, I therefore use four-digit HS codes, although the methodology can easily be applied at a more fine-grained level. At the four-digit level, I separate, for example, “Meat of swine” (0203) from “Meat of sheep and goats” (0204). Firms with more products than three standard deviations above the yearly mean are also removed.

I handle data breaks across years using concordance tables, since updated HS codes were introduced in 2012 and 2017. All HS codes are converted to their HS 2007 equivalents. I also transform all monetary values from both VARS and FIRM into real values using consumer price index (CPI) normalization to 2015, following Statistics Denmark⁷.

Appendix B.1 reports summary statistics for the sample. More than 3,900 unique firms are included, with an average of 72.7 employees. The average firm produces in 1.6 two-digit product categories, 2.3 four-digit product categories, and 2.9 six-digit product categories.

The distribution of firms across product scope, defined as the number of products manufactured, is highly skewed. A substantial number of firms produce only one product. Appendix B.1 furthermore reports the number and share of firms producing discrete numbers of products (at a four-digit level), from 1 to 10 or more. It also reports the average number of employees and occupations within each product scope category.

The number of manufacturers and the associated economic activity differ substantially across product categories. Table 7 in Appendix B.1 reports the top 10 products ranked by the number of firms registering sales in the product category and by average sales, respectively. There is some overlap across these rankings, while other products differ substantially in terms of the extensive and intensive margins of production. For instance, “7308: Iron or steel” is the product category with both the highest number of manufacturers and the highest yearly mean sales. “9403: Furniture” and “4418: Wood” also appear in both top 10 rankings. By contrast, products such as “7326: Iron or steel”, “3926: Plastics”, and “4911: Printed matter” are frequently present in firms’ product sets but have relatively low mean sales. “2309: Dog or cat food”, “9021: Orthopaedic or fracture appliances”, and “6810: Cement, concrete or artificial stone” are examples of products with relatively high sales and few

⁷<https://www.statistikbanken.dk/pris113>

manufacturers.

As expected, co-production patterns do not appear random either. Some bilateral product pairs are substantially more likely to be co-produced than others. The most frequently co-produced product pairs are “7204: Ferrous waste and scrap” in combination with “7308: Iron or steel” and “7326: Iron or steel”, which are co-produced by 81 and 78 firms, respectively. Five of the ten bilateral product pairs reported in Table 9 in Appendix B.1 belong to separate two-digit product categories. In addition to the two cases just mentioned, these include “7326: Iron or steel” with “7616: Aluminium”, and “3926: Plastics” with both “8480: Moulds” and “7326: Iron or steel”. Within the same two-digit product category, “9401: Seats” is frequently co-produced with “9403: Furniture”, and “4902: Newspapers, journals and periodicals” with “4911: Printed matter”, for example.

Taken together, these insights suggest that products differ in nature. For instance, “7204: Ferrous waste and scrap” appears in the two most frequent co-production pairings (cf. Table 9 in Appendix B.1) and is produced by many firms, ranking fourth in number of manufacturers too. However, as also shown in Table 8 in Appendix B.1, average per-producer sales in this product category are low, suggesting that it is a support product or by-product. Other products appear to be of a similar nature. For instance, “4911: Printed matter” is frequently co-produced with both “4901: Printed matter” and “4902: Newspapers, journals and periodicals”.

3.1.2 Workers and occupations

Information on workers in manufacturing firms is primarily drawn from the IDA databases, including IDAP, IDAN, and INDH. These data can be linked to the VARS data using a unique firm identifier and year. I can therefore assign each worker to one or more employers and observe changes in employment relationships. In the main sample, I keep only workers aged 25 to 60.

At the worker-year level, I observe occupational DISCO codes at the six-digit level. DISCO is the Danish equivalent of the ISCO classification and is intended to reflect the tasks performed by workers. Only worker-year observations with the highest DISCO quality

are kept in the sample. DISCO codes are grouped into nine main categories⁸, denoted by their first digit. These cover group 1: “Management”; group 2: “Work Requiring High-Level Skills, Including Research” (for instance medical doctors, engineers, and lawyers); group 3: “Work Requiring Medium-Level Skills” (for instance some functions within sales, business services, and administration); group 4: “Regular Office and Service Work” (for instance registration work); group 5: “Service and Sales” (for instance customer service roles); group 6: “Workers in Farming, Forestry, and Fishing” (including gardeners); group 7: “Artisanal Work” (including carpenters, construction workers, and workers in manufacturing, excluding machine operators); group 8: “Operator and Inspector Work”, including transportation and machine operators; and group 9: “Other Manual Work”, including cleaning, packing, and work from DISCO groups 6, 7, and 8 with no requirements for trained skills.

At more detailed levels, two-digit categories span 50 unique values. Selected examples include “21: Work in Science and Engineering”, “31: Technical Work in Science, Engineering, Shipping and Aviation”, and “81: Operator Work at Stationary Plants and Machines”. For the purposes of this paper, and especially given the inclusion of bilateral occupational skill-sharing weights, more detail is needed than the two-digit level provides. For instance, DISCO 21 benefits from being expanded, as “Work in Science and Engineering” is too broad for the present analysis.

At the three-digit level, two-digit code “21” expands into six sub-codes. These include “211: Physical/Earth Scientists”, “212: Math/Actuarial/Statistical Professionals”, and engineering occupations such as “214: Engineering Professionals” and “215: Electrotechnology Engineers”. Other examples include “311: Physical/Engineering Technicians” and “812: Metal Processing Operators”. In total, there are 167 unique three-digit DISCO codes in the sample.

At even more detailed levels, further sub-occupational categories are introduced. At the four-digit level, the DISCO variable covers 492 unique occupations. These four-digit codes expand the three-digit DISCO codes into further detail. For instance, “413: Text and Data Processing” is divided into two sub-occupations: “4131: Text processing” and “4132: Data registration work”. For the purposes of this paper, I regard this level of detail as unnecessary.

⁸A tenth main category, 0, covers military occupations.

The main sample therefore uses three-digit DISCO occupations.

A full list of DISCO codes across all aggregation levels can be found via Statistics Denmark⁹. By restricting the sample period to 2010 onward, I also avoid data breaks in the DISCO variable. All DISCO codes are therefore kept in the DISCO-08 version. An updated version was introduced in 2012, but the updates only affect DISCO codes at levels more detailed than four digits.

In total, 370,072 unique workers appear in the sample. Of these, 29 percent are female. The average worker appears in the sample for 4.42 years and registers in 1.25 distinct three-digit occupations. The average number of working hours is 1,412 per year, corresponding to 28.2 working hours per week under a 250-working-day calendar and a five-day working week. These statistics are summarized in Appendix B.2.

Some occupations are more frequently present across firms than others. In other words, some tasks are more commonly demanded by firms. Table 11 in Appendix B.2 displays the top 10 most prevalent occupations across all sample firms on the extensive margin. Column three and the corresponding share are constructed from firms reporting any presence of a given occupation across the sample period, and therefore do not capture the intensive margin of presence, such as hours worked or wage bill shares. Managerial occupations of different types (112, 132, 122, and 121, in ranked order of prevalence) are widespread across firms. In other one-digit occupational categories, presence is more heterogeneous across individual occupations. Some of the most common occupations are, for instance, “411: General Office Clerks”, “311: Physical/Engineering Technicians”, and “332: Sales/Purchasing Agents & Brokers”.

The extensive margin of occupations, also referred to as organizational complexity, differs substantially across firms. Appendix B.2 displays the kernel density of the distribution of firms by organizational complexity, measured as the number of unique occupations observed across the sample period. The observed distribution ranges from one to more than 50 occupations, although only very few firms register more than 40 occupations. The vast majority have fewer than 20 occupations, and the peak of the right-skewed distribution is below 10.

⁹<https://www.dst.dk/da/Statistik/dokumentation/nomenklaturer/disco-loen>

3.2 Product mix changes, occupational adding and dropping

When the number of products manufactured by a firm increases, the firm is likely to increase its demand for labor inputs. However, the type of labor demand increase is likely to depend on the nature of the firm's product adoption decisions. Here, I investigate correlation patterns in firm size and scope. I present three stylized facts. First, the average firm increases its product scope and number of employees over the sample period, but increases its organizational complexity to a lesser extent. Second, product scope is associated with more workers, but not more occupations, at the firm level when controlling for firm size. Third, a high product adding rate is associated with increased worker adding and with both increased occupational adding and dropping. The worker dropping rate does not increase significantly with either product adding or product dropping. Taken together, the first two stylized facts suggest that firms adjust their organization following product adding through targeted hiring. However, the third fact suggests that product adding is associated with both occupational adding and dropping, masking underlying occupational churn in the previous regressions.

3.2.1 Average firm scope and organization

The first stylized fact is motivated by a simple but revealing insight: the average firm grows in scope throughout the sample period and grows simultaneously in the number of employees and, to a lesser extent, the number of unique occupations. This pattern suggests that targeted hiring is taking place.

Figure 2 displays indexed growth in the sample average number of four-digit products, employees, and three-digit occupations. From 2010 to 2019, the number of products increases by approximately 12 percent, the number of employees increases by roughly 22 percent, whereas the number of unique occupations in the average firm increases by only 5 percent. Because employment grows faster than product scope, the average number of employees per product rises over the period, consistent with intensive-margin expansion within existing production lines. Notably, the number of products and employees increases steadily throughout the period, while the average number of occupations drops from 2013 to 2016 and does not recover

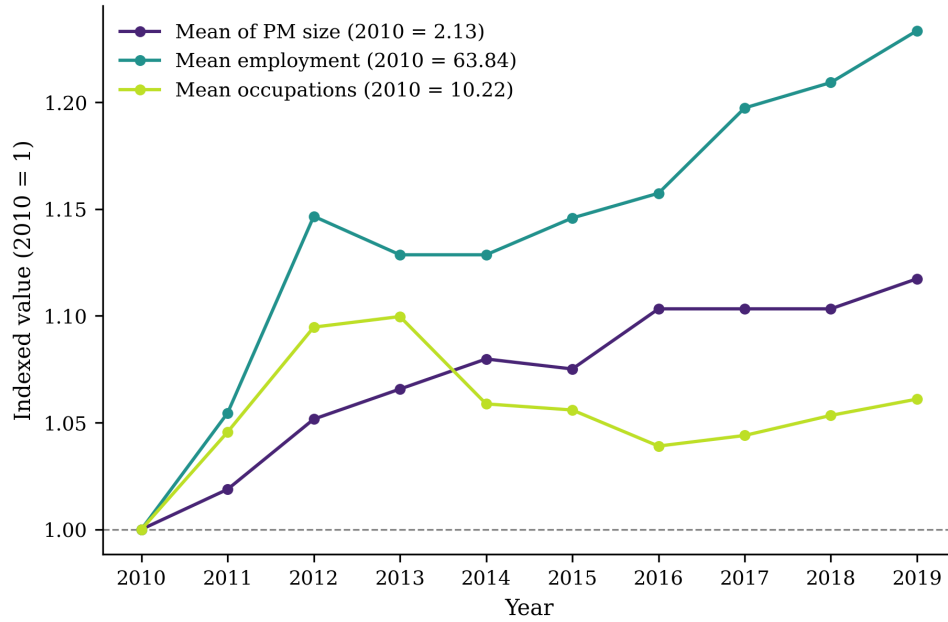


Figure 2: *This figure displays the average number of products, employees, and unique occupations in a sample firm across years. Products are grouped at the four-digit level, and occupations at the three-digit level.*

in the following years.

Further insight into yearly firm growth along these margins can be found in Appendix B.3, which displays growth figures in levels and at various aggregation levels, as well as alternative metrics. It shows that the firms grow in revenue and value added too, on average.

3.2.2 Workers and occupations in firm scale and scope

Next, I consider whether the association presented in Section 3.2.1 is consistent in firm-level data. As a first assessment of size and scope dependencies, I use linked employer-employee data covering all sample firms in the period. I regress the log number of employees and the log number of occupations, respectively, on the log number of products manufactured by the firm. The regression equation is

$$\log(y_{jt}) = \beta \log(|\Omega|_{jt}) + \lambda_j + \lambda_{s(j) \times t} + \varepsilon_{jt}, \quad (10)$$

Table 1: Employment and occupations in scale and scope

	log(employees _{jt})		log(occupations _{jt})	
	(1)	(2)	(3)	(4)
log(Ω_{jt})	0.0923*** (0.0257)	0.0551*** (0.0199)	0.0213 (0.0200)	0.0075 (0.0184)
log(revenue _{jt})		0.4055*** (0.0339)		0.1503*** (0.0213)
Firm FE	Yes	Yes	Yes	Yes
Sector $\times$ year FE	Yes	Yes	Yes	Yes
Observations	66,632	66,632	66,632	66,632
R ²	0.967	0.975	0.898	0.901

Notes: Clustered (firm) standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

where y_{jt} denotes either the number of employees or the number of occupations of firm j in year t , and $|\Omega_{jt}|$ denotes the size of the firm product mix (i.e. the number of products products). Hence, β measures the product-scope elasticity of employment and occupational breadth, respectively. No results should be interpreted as causal, as all variables are endogenous to the firm. In alternative specifications, I include $\log(\text{revenue}_{jt})$ as a direct control for firm scale.

As stated, I include firm fixed effects (FE). These absorb firm-specific factors influencing the number of products, including firm scale, managerial capabilities, time-invariant firm-specific dependencies in labor demand and technology, and geographical factors. Sector-year FE are further included to absorb common shocks to all firms in the same sector. These include macroeconomic factors, policy changes, sector-level technological changes, and, importantly, changes in common input prices and shared demand trends. The remaining variation is therefore intended to capture firm-specific changes that are unusual both within the firm and relative to other firms in the same sector.

The regression results suggest that broader product scope is associated with substantially larger employment, but not with broader occupational scope. Firms that expand their product portfolios therefore appear to scale primarily by adding more workers within a relatively stable set of occupations, rather than by proportionally widening the range of labor types they employ. This pattern is consistent with targeted hiring in incumbent occupations.

3.2.3 Product, worker, and occupational churning

Lastly, I consider whether the suggestive evidence for targeted hiring at the average and firm levels, shown in Sections 3.2.1 and 3.2.2, respectively, masks an underlying churning mechanism. For instance, if firms reorganize by simultaneously adding and dropping occupations, this will be masked in the regression reported in Table 1.

I define a churn rate as the sum of additions and separations relative to the size of the stock:

$$\text{churn rate} = \text{add rate} + \text{drop rate} = \frac{\# \text{additions} + \# \text{separations}}{\text{stock}},$$

which I consider for products, workers (by unique worker ID), and occupations. Columns (1) and (4) in Table 2 display the estimation results from

$$Y_{jt} = \beta \text{Product churn rate}_{jt} + \lambda_j + \lambda_{s(j) \times t} + \varepsilon_{jt}, \quad (11)$$

where Y_{jt} is either the worker churn rate or the occupation churn rate. The results suggest that an increase in product churn is not associated with an increase in worker churn, but is significantly associated with more occupational churning.

Because the aggregate churn rate pools additions and separations, it can mask offsetting associations across the two margins; I therefore decompose it into adding and dropping. In columns (2), (3), (5), and (6), I estimate

$$Y_{jt} = \beta X_{jt} + \lambda_j + \lambda_{s(j) \times t} + \varepsilon_{jt}, \quad (12)$$

where Y_{jt} is the worker add rate, worker drop rate, occupation add rate, or occupation drop rate, and X_{jt} is either the product add rate or the product drop rate.

The results suggest that a higher product adding rate is associated with a higher worker adding rate, however, *not* with more worker dropping. Product adding is similarly associated with an increase in occupational adding, but contrary to worker dropping, it is significantly associated with more occupational dropping. At the organizational level, this pattern can be interpreted to suggest targeted reorganization: firms need more workers in *some* occupations and need *some* new occupations, and they fill these positions using both new hires and

Table 2: Churn regressions

	Workers			Occupations		
	Churn rate (1)	Add rate (2)	Drop rate (3)	Churn rate (4)	Add rate (5)	Drop rate (6)
Product churn rate	0.0329 (0.1301)			0.0781*** (0.0270)		
Product add rate		0.0414*** (0.0152)	0.2363 (0.2650)		0.0446*** (0.0146)	0.1570** (0.0625)
Product drop rate		0.0139 (0.0095)	-0.0962 (0.1179)		0.0149 (0.0093)	-0.0010 (0.0171)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	56,877	56,877	56,877	56,877	56,877	56,877
R ²	0.192	0.412	0.191	0.312	0.294	0.296

Notes: Clustered (firm) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

incumbent workers. Simultaneously, some occupations become redundant, for instance due to task reallocation. Increased product dropping is not associated with either increased or decreased worker adding, worker dropping, occupational adding, or occupational dropping.

Taken together, the stylized facts point to a consistent pattern of targeted hiring and re-organization in response to changes in product scope. Firms that expand their product portfolios primarily increase employment within existing occupations, while only marginally expanding the set of occupations they employ. At the same time, product adding is associated with both occupational entry and exit, indicating ongoing reallocation of tasks within firms rather than broad-based expansion of capabilities. These patterns suggest that firms rely heavily on their existing workforce structure when adjusting their product mix. Combined with the idea that firms endogenously choose products from a set of potential candidates that are the most similar to their core competencies (cf. Section 2), this evidence motivates the central conjecture that the paper goes on to test: that firms select products closer in task space to their current capabilities and exploit the ability to reallocate tasks more efficiently when scope grows. In other words, product expansion appears to follow paths that economize on existing occupational structures, supporting the view that endogenous product mix adjustment decisions are guided by exploitation of flexible manufacturing.

4 Required occupations to produce

Section 2 presented a model in which firms guide their product adoption decisions toward candidate products that require similar task inputs to the firm’s incumbent core product, and Section 3.2 presented stylized facts to support both the idea of targeted hiring and reorganization from product scope expansions. To move forward, I need an empirical proxy for the task similarity of products, denoted $\Phi(\omega, \omega')$.

In this section, I construct a product-level measure of the occupations required for production, and hence implicitly the underlying tasks, across the product space. The methodology outlined in the introduction relies on using occupational structures of firms producing within a single product category to proxy for product-level task requirements. This requires a binary classification of firms into SPFs and MPFs.

Appendix C.1 contains an elaborate data-driven approach to this. It shows that many firms produce only one product, but they cover only a small share of the product space, introducing a trade-off between product coverage and cleanliness of the associated occupational vectors. I show that by allowing for a small share of co-production outside of the core product, SPFs cover a larger share of the product space, and the vast majority of SPF sales are still found in the core product.

Going forward, I treat SPFs as firms with 90 percent or more of their manufacturing sales in a single product category in every sample year.¹⁰ Under this definition, the SPF-covered set $\mathcal{P}$ spans 44 percent of four-digit product categories, which together account for nearly 93 percent of total manufacturing sales over the period. Within SPFs, nearly 99 percent of sales still come from the core product.

In the following section, I outline the grouping method for constructing product-level occupational vectors from SPFs, and I illustrate the content of the data.

¹⁰By extension, MPFs are defined as having less than 90 percent of their sales in the core product across all sample years. Therefore, I also exclude firms that switch status across the sample period. MPFs in the sample produce 6.4 products on average. The average sales-share concentration in the core product is 63 percent, and the mean sales share across all products is 26 percent.

4.1 Product-level vectors of occupations

Using the occupational data described above and the SPF definition provided in Appendix C.1, I construct product-level vectors of required occupations, denoted $\mathcal{O}(\omega)$ for all $\omega \in \mathcal{P}$, where $\mathcal{P}$ denotes the full product space covered by SPFs. A key step is to move from the extensive margin of occupational presence, considered in Section 3.2, to the intensive margin. The objective is therefore to measure the relative importance of each occupation in the production of a given product.

I treat the required occupational inputs as static, as indicated by the absence of a time subscript t . This reflects a practical limitation of the data: the set of SPFs is thin in any single year, so the SPF-covered product space $\mathcal{P}$ and the associated vectors would be noisy and unbalanced if constructed year by year. The downside of omitting the time dimension is sensitivity to time-varying changes in production technologies that affect task requirements at the product level. In all relevant regressions, I attempt to mitigate this through interacted fixed effects.

For each firm in the set of SPFs, I first compute the total number of hours worked in each observed occupation. Let SPF denote the full set of single-product firms. Firms are then grouped by their core product $\bar{\omega}_j$, such that each $j \in \text{SPF}$ belongs to the set of SPFs producing $\omega = \bar{\omega}_j$. I denote the set of SPFs producing product ω by $\text{SPF}_\omega \subseteq \text{SPF}$.

Occupation-specific working hours at the product level are computed as the sum of hours worked in occupation o across all firms $j \in \text{SPF}_\omega$. Formally,

$$h_{\omega o} = \sum_{j \in \text{SPF}_\omega} h_{jo}, \quad \text{and} \quad h_\omega = \sum_o h_{\omega o}.$$

The relative importance of occupation o in producing product ω is then given by the share of total labor hours:

$$\theta_{\omega o} = \frac{h_{\omega o}}{h_\omega}, \quad \text{with} \quad \sum_o \theta_{\omega o} = 1.$$

This yields one product-level vector $\mathcal{O}(\omega)$ per covered product—that is, $|\mathcal{P}|$ vectors in total—each of length N_O , where N_O denotes the total number of unique occupations. Each vector

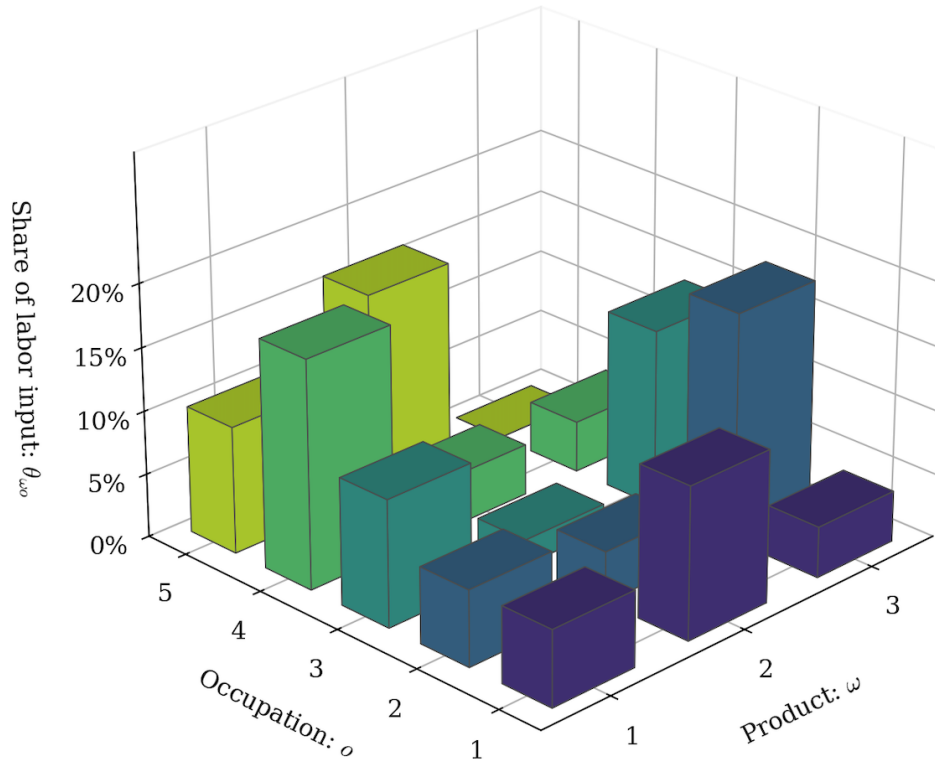


Figure 3: *Figure 3 illustrates product-level occupational vectors for three hypothetical products, each using five occupations. The height of each column reflects the share of labor input hours. All data are fictional and constructed for illustrative purposes.*

summarizes the occupational composition of production for a given product:

$$\mathcal{O}(\omega) = (\theta_{\omega 1}, \theta_{\omega 2}, \dots, \theta_{\omega N_O}).$$

Under the assumption that occupations proxy for the underlying tasks, these vectors can be interpreted as reduced-form representations of product-specific production technologies in task space.

Figure 3 illustrates such vectors for three hypothetical products with five occupational inputs. Products and occupations are represented along the horizontal dimensions, while the height of each column reflects the relative importance of each occupation, measured as the share of labor input hours.

In this example, occupation 1 (e.g., managers) is used in the production of all products, but is most important for product 2. Occupation 2 (e.g., specialists) is also used across all

products, but most intensively in product 3. Occupation 3 (e.g., office work) is primarily used in products 1 and 3, but is also present in product 2. Occupation 4 (e.g., sales roles) is most important for product 1, but contributes to production across all products. Finally, occupation 5 (e.g., assembly work) is used only in products 1 and 2 and is absent from product 3. Numerical values for this example are provided in Appendix D.1.

5 Product-to-product similarities in occupations

In this section, I show how to map similarities between products based on their occupational vectors, building on structures such as those illustrated in Figure 3.

Construction of task similarities between all bilateral product pairs, $\Phi(\omega, \omega')$, relies on cosine similarity between the associated occupational vectors, following a similar approach to Boehm et al. (2022), Bloom et al. (2013), and others. Formally, task relatedness between two products is defined as the inner product of their occupational vectors, $\langle \mathcal{O}(\omega), \mathcal{O}(\omega') \rangle$.

Using a standard cosine similarity measure, each occupational input share in ω is compared only to the corresponding occupation in ω' . However, not all occupations are equally distinct in terms of the tasks they perform. A simple inner product therefore does not account for heterogeneity in skill or task overlap across occupations. To address this, I introduce a weighting matrix Ψ , with element Ψ_{o,o^*} capturing the estimated skill transferability from an origin occupation o to a destination occupation o^* .

With this extension, occupations are no longer compared only one-to-one, but each occupation is compared to all other occupations, weighted by their degree of estimated skill relatedness. As a result, all bilateral occupation pairs can contribute to product similarity, provided that both occupational presence and skill overlap are sufficiently large. The magnitude of task similarity is therefore determined jointly by the relative importance of occupations and the degree of estimated skill transferability between them. The resulting similarity measure between products ω and ω' is defined as:

$$\Phi(\omega, \omega') = \frac{\sum_{o,o^*} \Psi_{o,o^*} \cdot \theta_{\omega o} \cdot \theta_{\omega' o^*}}{\sqrt{\sum_{o,o^*} \Psi_{o,o^*} \cdot \theta_{\omega o} \cdot \theta_{\omega o^*}} \cdot \sqrt{\sum_{o,o^*} \Psi_{o,o^*} \cdot \theta_{\omega' o} \cdot \theta_{\omega' o^*}}} \quad (13)$$

where $\theta_{\omega o}$ denotes the share of working hours assigned to occupation o among all SPFs producing product ω , following Section 4, such that each element corresponds to the height of the columns in Figure 3. The weighting matrix Ψ is square, with $\Psi \in \mathbb{R}^{N_O \times N_O}$, where N_O denotes the number of unique occupations, equal to 167 at the three-digit level.

Appendix D.1 provides a numerical example of both unweighted and weighted cosine similarities for the products illustrated in Figure 3. The example shows that introducing the

weighting matrix not only affects the magnitude of bilateral similarities, but can also alter the ranking of similarity across product pairs.

The following section describes the construction of the weighting matrix. I then compute and present bilateral product similarities based on Equation (13) across the full product space.

5.1 Bilateral occupation skill-relatedness

The skill-relatedness matrix, Ψ , is constructed following the approach of Neffke and Henning (2013), who estimate skill relatedness between industries using excess labor flows. I adapt this method to the occupational level, in line with Joyez and Laffineur (2022), who construct an “Occupation Space” using French data. Specifically, each element of the matrix is defined as:

$$\Psi_{o,o^*} = \frac{F_{o,o^*}}{\hat{F}_{o,o^*}}, \quad (14)$$

where F_{o,o^*} denotes the observed worker flow from an origin occupation o to a destination occupation o^* , and $\hat{F}_{o,o^*}$ is the expected flow, estimated using a zero-inflated negative binomial (ZINB) regression with occupation-level characteristics as predictors.

Formally, the observed flow is defined as the number of workers who change occupational code from an origin occupation o to a destination occupation o^* during the sample period. Importantly, I do not restrict the sample of worker flows to those employed in manufacturing firms, as the estimation of bilateral occupation skill relatedness is independent of product-level data. Groes et al. (2015) report that close to one fifth of workers in Denmark change occupations within a year, which is slightly higher than the figures reported by Humlum (2021b) and Humlum (2021a). Here, a little more than 10 percent shift occupations in a given year after 2011. Similar magnitudes are found for the United States by Moscarini and Thomsson (2007) and Kambourov and Manovskii (2008). This is more in line with the magnitude of job switches in my sample. Appendix D.2 reports that between 115 thousand and 153 thousand workers switch occupations each year, corresponding to between 7.8 and 10.3 percent of the worker sample.

Within particular occupation codes, there is asymmetry between inflows and outflows of

workers, which supports the appropriateness of an asymmetric specification of Ψ_{o,o^*} . Appendix D.3 illustrates observed worker in- and outflows for selected occupations.

I do not distinguish between job switchers and workers who change occupational codes within ongoing employment relationships. Groes et al. (2015) show that occupational transitions are more frequent among job switchers, but a substantial share of transitions also occur within firms. This suggests that internal labor markets play an important role in generating occupational flows. In the present data, approximately 61 percent of occupational changes occur in relation to a job switch.¹¹ Around half of the occupation changes move upward across the numerical DISCO categories, and the remainder move downward (as displayed in Appendix D.4).

At the level of bilateral occupation pairs, observed flows are sparse. Approximately half of all occupation pairs exhibit zero flows, while nearly ten percent of pairs involve only a single worker transition. Appendix D.5 presents a binned histogram of bilateral occupation pairs by flow count. These patterns support the use of a model that explicitly accounts for zero-inflation. The ZINB specification is particularly suitable, as it jointly models (i) the probability that a positive flow occurs and (ii) the size of the flow, capturing both the extensive and intensive margins of skill transferability.

Estimation results are presented in Appendix D.6. Higher average wages, wage growth rates, and employment growth in a destination occupation are associated with larger inflows. In addition, occupations with larger employment levels tend to receive more workers. I estimate that all occupations with sufficiently large employment experience positive inflows over the sample period.

The occupation pairs with the highest estimated skill transferability are reported in Table 3. Occupations within the same main categories (see Section 3.1) often exhibit high skill relatedness. For instance, “232: Vocational Education Teachers” has a high estimated skill relatedness with “233: Secondary Education Teachers”; “713: Painters/Building Cleaners” with “712: Building Finishers”; and “721: Metal Workers/Welders” with “722: Blacksmiths/Toolmakers”. However, many occupations are also estimated to share required skills with

¹¹An occupation change for a job switcher is defined by a simultaneous change in employer ID and occupation code for the worker from $t - 1$ to t .

Table 3: Top occupation pairs by skill sharing.

Origin o	Destination o^*	F_{o,o^*}	Ψ_{o,o^*}
512: Cooks	343: Arts/Cultural/Culinary Assoc.	1219	152.8016
232: Vocational Educ. Teachers	233: Secondary Educ. Teachers	2168	99.7396
512: Cooks	941: Food Prep. Assistants	1803	98.6495
713: Painters/Building Cleaners	712: Building Finishers	1908	85.8258
541: Protective Service Workers	325: Other Health Associates	2009	82.7434
224: Paramedical Practitioners	541: Protective Service Workers	91	76.5060
252: Database/Network Prof.	351: ICT Ops/User Support	945	74.9422
612: Animal Producers	324: Veterinary Techs/Assistants	47	68.4551
721: Metal Workers/Welders	722: Blacksmiths/Toolmakers	2061	67.9408
941: Food Prep. Assistants	512: Cooks	1615	63.4599

Notes: This table contains the top 10 bilateral occupation pairs in terms of estimated excess skill-transferability from any origin occupation o to a destination occupation o^* . For each bilateral pair, the table contains the observed flow F_{o,o^*} and the excess flow estimation Ψ_{o,o^*} following Equation 14, which is controlled for occupation-specific size and wage characteristics potentially driving flows. The expected flow is estimated using a ZINB model accounting for a large share of observed 0s. All text labels of occupations are generated using Anthropic’s Claude from author provided text descriptions in Danish, prompted to translate and shorten occupational titles for readability purposes.

occupations outside of their one-digit group. For instance, “512: Cooks” have a high excess worker flow to “343: Arts/Cultural/Culinary Associates” (which includes “3434: Kitchen Chef”) and to “941: Food Preparation Assistants”, as both an origin and destination occupation. “541: Protective Service Workers” (which includes “5419: Ambulance Rescuers” among other relevant four-digit occupations) relate highly to “325: Other Health Associates” and “224: Paramedical Practitioners” (the latter as destination).

In general, some occupations entail more transferable skills than others. Appendix ?? displays the top 10 occupations o with the highest mean skill relatedness $\bar{\Psi}_o$ to all possible destination occupations o^* . The table reports both $\bar{\Psi}_o$ and the number of observed destination occupations. “224: Paramedical Practitioners” has the highest mean skill relatedness but relatively few observed destinations. Occupations “512: Cooks” and “343: Arts/Cultural/Culinary Associates” are also highly generic in skills and have more observed destination occupations. Particularly relevant for the sample used, “821: Assemblers” is highly generic, as are both “818: Other Plant/Machine Operators” and “721: Metal Workers/Welders”, among other manufacturing-oriented jobs.

5.2 Visual depiction and product clusters

The product relatedness estimates derived from Equation (13) are visualized as a symmetric similarity matrix, with products on both axes. In the heatmap shown in Figure 4, brighter yellow colors indicate higher occupational similarity between product pairs, while darker blue colors indicate lower similarity.

For confidentiality reasons, not all products are included in the figure. Specifically, the matrix displays only products with more than five unique SPFs observed over the period 2010–2019 (among other restrictions), i.e., products with $|\text{SPF}_\omega| > 5$, in line with GDPR requirements. This constraint leads to the exclusion of a disproportionately large number of numerically low product categories.

The heatmap reveals several clear clusters in the product space. First, the upper-left corner contains a compact food-related cluster, with strong similarities among “0201: Meat (fresh/chilled)”, “0304: Fish (fillets/fish meat)”, “1601: Meat preparations”, “1604: Fish preparations”, and “1704: Sugar confectionery”. Within this cluster, pairwise similarity values are high. For example, the similarity between “1601: Meat preparations” and “1604: Fish preparations” is approximately 0.98, while “0201: Meat (fresh/chilled)” is also highly similar to both (around 0.97 and 0.96, respectively).

Second, there is a cluster of plastic and rubber products centered around “3920: Plastics” and “4016: Rubber”. This cluster also includes “3923: Plastics and Ethylene polymers”, “3925: Plastics”, and “3926: Plastics”. Strong links are observed, for instance, between 3920 and “3926: Plastics”, and between 3920 and “4016: Rubber”.

The most prominent feature of the heatmap is a large machinery and equipment cluster located in the central part of the matrix, spanning roughly from “8412: Engines” to “8486: Machines and apparatus”. The densest core includes products such as “8421: Centrifuges and machinery”, “8422: Dishwashing machines and machinery”, “8424: Fire extinguishers and spray guns”, “8426: Cranes and derricks”, “8428: Lifts, elevators and conveyors”, and “8431: Machinery and lifts”. This cluster forms the largest contiguous region of high similarity, suggesting a shared underlying capability structure. It should be noted that GDPR-based inclusion criteria disproportionately favor this region of the product space.

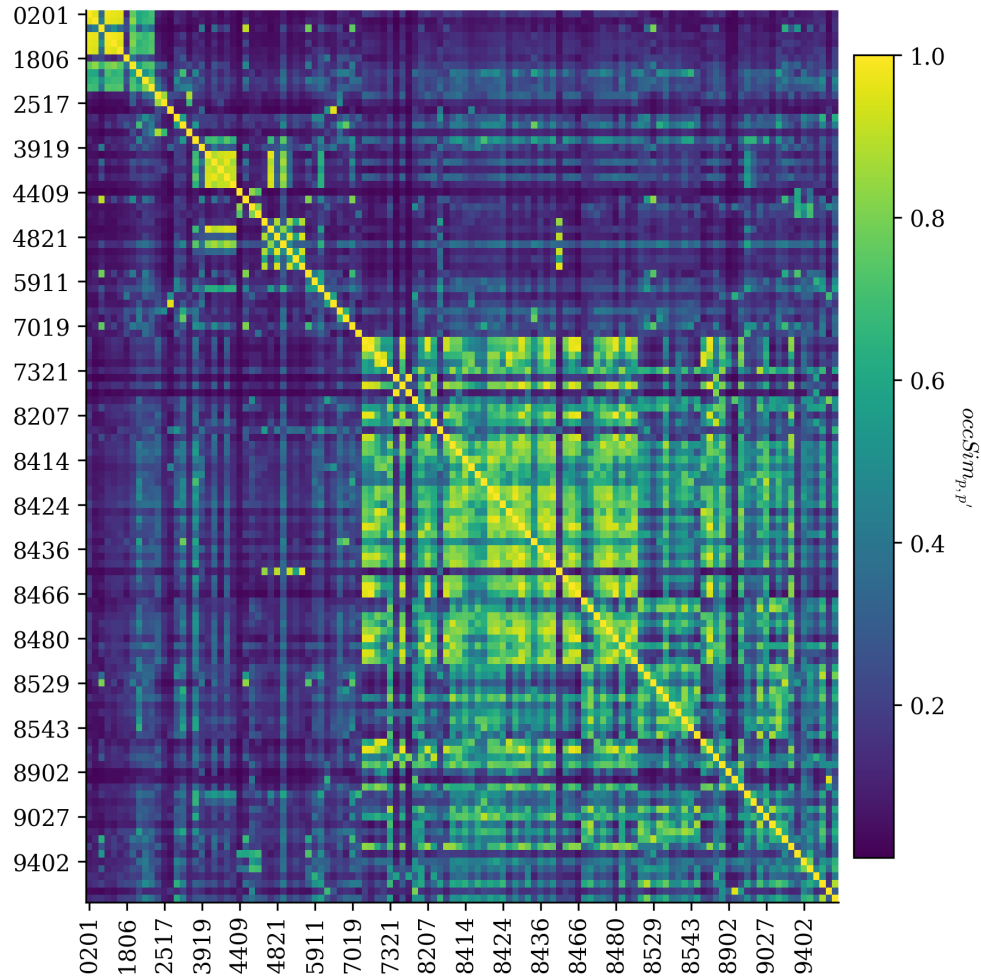


Figure 4: *Figure 4 displays product similarities in occupations, $\Phi(\omega, \omega')$, constructed from Equation 13. High similarities (closer to 1) appear in brighter yellow, while lower similarities (closer to 0) appear in darker blue. The diagonal contains values equal to 1. Only products satisfying GDPR requirements are included.*

An important feature of the figure is that high similarities do not necessarily occur between numerically adjacent product codes. For example, the occupational composition of “8707: Vehicles” differs substantially from that of “8708: Vehicle parts and accessories”, despite their proximity in the classification. In contrast, “8707” is more similar to products such as “8433: Mowers and haymaking machinery” or “7308: Iron or steel”.

Other notable examples include “8207: Tools, interchangeable”, which is closely related to “8480: Moulds” and “8466: Machine-tools” (both around 0.98). Similarly, “7304: Iron or steel (excluding cast iron)” is highly similar to “8480”, “8466”, and “8707”. “7326: Iron

or steel” aligns more strongly with “8431: Machinery and lifts” and “8437: Machines and machinery” than with many nearby codes.

Some isolated pairings stand out particularly clearly: “4911: Printed matter” is highly similar to “8442: Machinery, apparatus and equipment” (approximately 0.96), “7309: Reservoirs, tanks, vats and similar containers” to “8609: Containers” (approximately 0.96), and “9406: Buildings prefabricated” to “8903: Yachts and other vessels” (approximately 0.95). These off-diagonal relationships suggest that the similarity measure captures shared production and occupational requirements that cut across the HS classification, rather than simply reflecting proximity in product codes. Appendix D.7 reports the 30 most similar bilateral product pairs, which confirms the patterns observed in the heatmap.

More generally, high similarities are relatively rare, as most product pairs share limited overlap in occupational composition. Appendix D.8 presents the distribution of bilateral product similarities. The distribution is right-skewed and peaks around 0.2, while only a small fraction of product pairs exhibit similarity values above 0.8.

5.3 Assumptions and robustness

The main results on bilateral product task similarities rely on a number of assumptions and methodological choices by the author. In this section, I address several of the most important ones and provide brief robustness analyses which aim to inform how the task similarity measure changes with sample or measurement alterations. Table 5.3 contains correlation metrics relating the benchmark task similarity matrix illustrated in Figure 4 to each alternative construction, including Pearson r , Spearman ρ , and Kendall τ . The former measures linear correlation, and is thus indicative of whether two matrices assign the same relatedness *levels* to product pairs, whereas Spearman ρ and Kendall τ instead assess whether the two constructions agree on the *ordering* of which products are close in task space. Because ρ and τ depend only on ordering, they are invariant to any monotone rescaling of similarity. This scale invariance is particularly important for the product selection mechanism motivated in Section 2, since that mechanism operates on which candidates rank closest in task space rather than on cardinal similarity; τ provides the more conservative, tie-robust reading of that ordinal agreement. Note that τ tends to be lower than r for a given strong monotone

association, so the gap between the two across the high-correlation rows below is partly mechanical rather than evidence of reordering.

A first assessment concerns the role of the weighting matrix, corresponding to the standard cosine similarity approach used in the literature (for input-based similarities, see Boehm et al. (2022); for sector-level similarities, see Bloom et al. (2013)). Correlating their off-diagonal elements yields a Pearson r of 0.974, while the rank-based coefficients are somewhat lower ($\rho = 0.962$, $\tau = 0.860$). Omitting Ψ_{o,o^*} thus leaves the levels of product relatedness essentially intact and perturbs mainly their ordering. Since the selection mechanism operates on ranks, this ordinal movement—rather than the near-unchanged levels—is the economically relevant margin along which the weighting matrix matters, even though the measure is otherwise highly robust to its omission.

A second alternative replaces the hour-weighted occupational vector with a vector of extensive-margin binary indicators of occupational presence among any SPF_ω , thereby assessing the importance of the hour-weighting in constructing $\mathcal{O}(\omega)$. One can think of this specification as a two-dimensional version of Figure 3, in which the share of labor input is uniformly distributed. Here both level and rank correlations with the benchmark are low ($r = 0.361$, $\tau = 0.300$), indicating that the intensive-margin hour weights in $\mathcal{O}(\omega)$ carry substantial economic content: they shape *which* products read as task-proximate, not merely the scale of similarity. Adding Ψ_{o,o^*} to this extensive-margin construction barely moves the correlations (0.369 versus 0.361 in r).

In constructing the product-level occupational vectors, all occupations are included. Ideally, one might distinguish between workers directly involved in production and those performing administrative or managerial tasks. This gives rise to an implicit assumption: all occupations contribute to production in proportion to their employment shares. Addressing this concern empirically is challenging. One possible, but rather crude, approach is to exclude managerial occupations (DISCO group 1) when constructing the product-level vectors. The off-diagonal Pearson correlation with the benchmark is 0.952, the rank-based coefficients only somewhat lower ($\rho = 0.922$, $\tau = 0.789$). The relatedness pattern is therefore driven by employment outside the managerial occupations, consistent with production- and task-specific labor, rather than administrative composition, accounting for cross-product proximity.

Table 4: Correlation between matrix measures

Matrix	Pearson r	Spearman ρ	Kendall τ
Unweighted cosine	0.974	0.962	0.860
Extensive margin occupational presence	0.361	0.430	0.300
Unweighted cosine, extensive margin	0.369	0.432	0.302
No managers	0.952	0.922	0.789
Inverse productivity weighting	0.985	0.984	0.916
Input	0.112	0.151	0.103

Notes: Each row reports the association between two matrices based on their off-diagonal correlations. *Pearson* r measures the linear correlation between corresponding entries. *Spearman* ρ is the rank correlation, capturing any monotonic (not necessarily linear) association. *Kendall* τ is a rank correlation based on the share of concordant minus discordant pairs, and is typically more conservative in magnitude than ρ . 'Unweighted cosine' reports the similarity matrix not including Ψ_{o,o^*} : $\langle \mathcal{O}(\omega), \mathcal{O}(\omega') \rangle$; 'Extensive margin occ. presence' disregards the intensive margin presence and only considers binary presence; 'Unweighted cosine, extensive margin' disregards both hour-weighting and Ψ_{o,o^*} ; 'No managers' disregards all occupations in main category 1; 'Inverse productivity weighting' weights SPFs within products by their inverse productivity measured by VA per worker; 'Input' computes the similar matrix using sourcing information (not all products covered by input matrix). All coefficients are statistically significantly different from 0 at the 1% level ($p < 0.001$).

A further concern relates to the underlying distribution of SPF vectors within products. Even if two firms employ the same occupations, workers may not perform identical tasks across firms, which raises the question of whether occupations accurately map into task requirements. Some firms may be more likely to assign workers to tasks that differ from their formal occupation codes. I perform a brief robustness exercise in which product-level vectors are weighted by the inverse of value added per worker of the SPF, a proxy for firm-level productivity. The premise is that more productive firms have greater flexibility in assigning workers outside their core tasks and should therefore receive less weight in the construction of product-level vectors. The reweighted matrix is almost indistinguishable from the benchmark ($r = 0.985$, $\tau = 0.916$). The concern therefore appears quantitatively minor.

Understanding occupational similarity between products is also improved when related to alternative measures. One obvious comparison is product relatedness in alternative inputs, for instance via firms' sourcing of intermediate inputs. I compute an analogous matrix using the VARK database for all SPFs in the sample, grouping SPFs by their product and constructing a vector of sourcing inputs at the four-digit product level. I then take the

(unweighted) cosine similarity between these vectors to construct the ‘Input’ relatedness matrix. This input-based matrix correlates only weakly with the occupation-based benchmark, in both levels and ranks ($r = 0.112$, $\tau = 0.103$). Occupational task proximity and intermediate-input proximity therefore capture largely distinct dimensions of product relatedness: firms close in what they source are not, in general, close in the labor capabilities their production requires.

6 Multi product firms and occupational similarity

In this section, I examine how occupational similarity shapes product scope decisions of MPFs, in line with the theoretical predictions outlined in Section 2. The central mechanism is that firms expand and contract their product mix along dimensions of task similarity to their core competencies.

I begin by documenting how product scope within firms is organized relative to task similarity. MPFs structure their product portfolios in a way consistent with the framework of Eckel and Neary (2010): products that are closer in task space to the firm's core product tend to play a more prominent role. Figure 5 illustrates this relationship by plotting the average task similarity of products to the firm's core product as a function of product rank (ordered by sales shares).

The figure reveals a clear negative relationship. Products with lower sales shares are systematically less similar to the firm's core product. This pattern suggests that firms concentrate production in products that are closest to their existing capabilities, while more distant products play a more marginal role. In terms of Section 2, this is the empirical counterpart of the viable range $[\bar{\omega}_j, \underline{\omega}_j]$: similarity to the core declines toward the marginal product $\underline{\omega}_j$ as rank falls. Although product-level productivity is not directly observed, this evidence indicates that production scale is increasing in task similarity to core competencies.

I next turn to product scope adjustments, and start by showing that product adding decisions are significantly directed toward products that share tasks with the firm's core competencies. To quantify this relationship, I estimate the probability of adding a given product as a function of its task similarity to the firm's core product:

$$P(\text{Add}_{j\omega,t}) = \beta \Phi(\bar{\omega}_j, \omega) + \lambda_j + \lambda_t + \lambda_{\bar{\omega}_j} + \lambda_{\omega} + \varepsilon_{j\omega,t}. \quad (15)$$

The specification includes firm fixed effects, year fixed effects, core-product fixed effects, and candidate-product fixed effects. These absorb firm-specific propensities to expand, time-specific shocks, heterogeneity in core-product characteristics, and differences in the general attractiveness of candidate products, respectively.

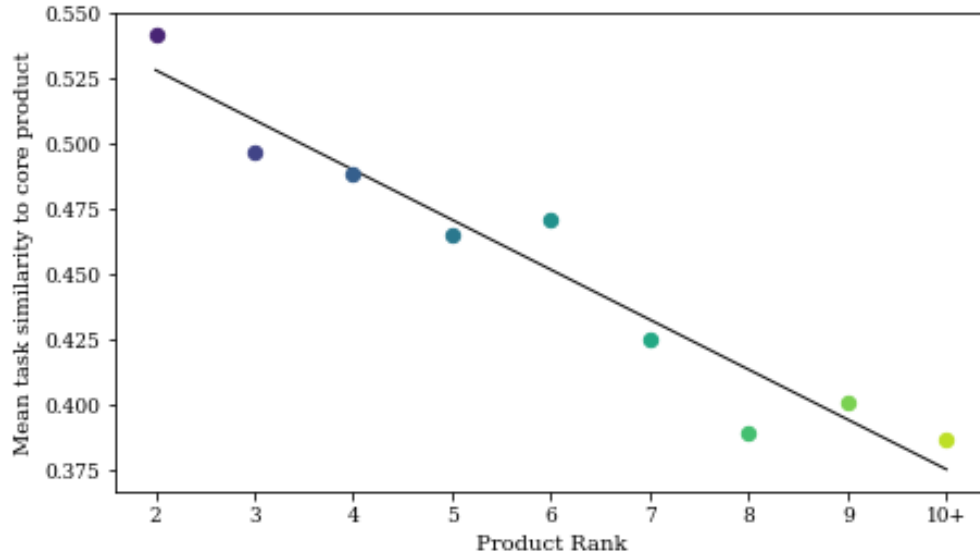


Figure 5: *This figure displays the mean task similarity (across firms and years) of co-produced products by rank. Since MPFs vary in product scope, the number of firms contributing to the mean decreases with rank. The black line shows a linear least squares fit.*

The results, reported in Table 6, show that a one-unit increase in task similarity increases the probability of product addition by 1.3 percentage points. Since $\Phi \in [0, 1]$, this one-unit change spans the entire similarity range, and the magnitude should be interpreted relative to a very low baseline probability of product addition, as the unit of observation is at the firm–year–product level across the full product space. Across the roughly 0.2 similarity range spanned by within-firm products in Figure 5, the linear estimate implies only about 0.3 percentage points—small relative to the top-percentile effect documented below, which already foreshadows the nonlinearity in adding behavior.

Importantly, this finding provides direct empirical support for the theoretical mechanism: firms expand along dimensions where marginal production costs are lower due to overlap in task requirements.

Turning to product dropping decisions, I estimate an analogous specification, with the outcome measured in $t + 1$ and all regressors in t :

$$P(\text{Drop}_{j\omega,t+1}) = \beta \Phi(\bar{\omega}_j, \omega) + \lambda_j + \lambda_t + \lambda_{\bar{\omega}_j} + \lambda_{\omega} + \varepsilon_{j\omega,t+1}. \quad (16)$$

The results, also reported in Table 6, indicate a strong negative relationship between task

Table 5: Linear estimates of product adding and dropping

	Add _{jω,t}	Drop _{jω,t+1}
$\Phi(\bar{\omega}_j, \omega)$	0.0133*** (0.0003)	-0.0630*** (0.0066)
<i>Fixed effects</i>		
Firm j	Yes	Yes
Year t	Yes	Yes
Core product $\bar{\omega}_j$	Yes	Yes
(candidate) Product ω	Yes	Yes
Observations	4,266,100	20,202
R ²	0.010	0.445
Within R ²	0.0043	0.0031

Notes: Clustered (firm) standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

similarity and product dropping. A one-unit increase in similarity reduces the probability of dropping a product by 6.3 percentage points. This effect is economically large. For reference, Figure 5 shows that the difference in similarity between highly ranked and low-ranked products is approximately 0.2, implying a substantial change in dropping probabilities across the product distribution.

Taken together, the adding and dropping results show that firms adjust their product portfolios asymmetrically: they expand into products that are similar to their core competencies and exit products that are more distant in task space. This is consistent with firms continuously reallocating production toward their most efficient uses of labor.

While the baseline regressions impose a linear relationship, the adapted framework of Section 2 predicts that the adding probability is a nonlinear function of similarity whose curvature is governed by the cost-curvature parameter η . To recover this shape, I estimate a percentile-based specification, in which candidate products are grouped according to their similarity to the firm's core product. Equation (17) states the specification:

$$P(\text{Add}_{j\omega,t}) = \sum_{p=2}^{100} \beta_p \mathbf{1}\{\Phi(\bar{\omega}_j, \omega) \in \mathcal{Q}_p\} + \lambda_j + \lambda_t + \lambda_{\bar{\omega}_j} + \lambda_{\omega} + \varepsilon_{j\omega,t}, \quad (17)$$

where $\mathcal{Q}_p$ denotes the p -th percentile bin and the lowest bin is the omitted reference. Fixed effects are included to match the main specification of Equation (15). The results are pre-

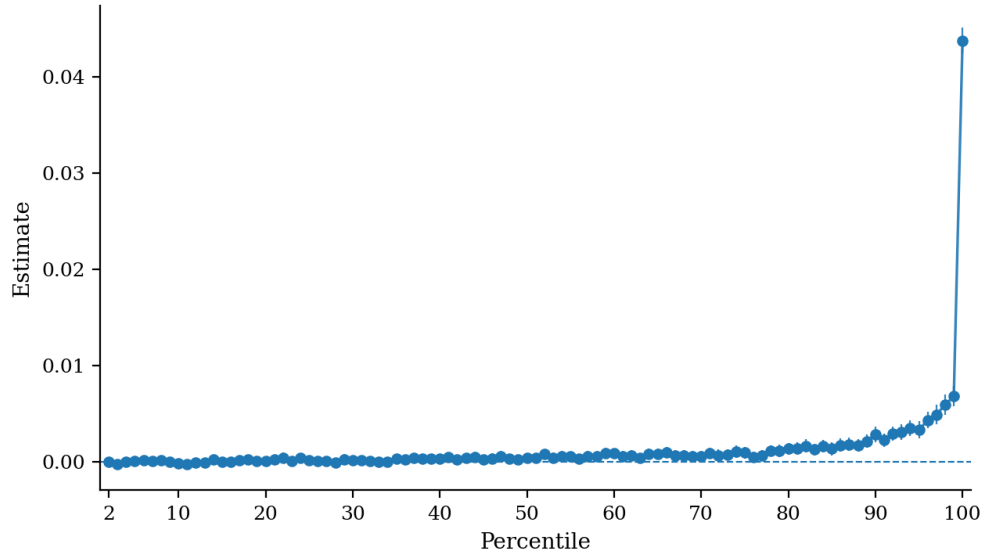


Figure 6: *This figure displays the probability of product addition by percentile of task similarity to the core product.*

sented in Figure 6, where the y -axis displays the β_p coefficients and the x -axis denotes the percentile p .

The results reveal a highly nonlinear relationship. Product addition is concentrated almost entirely among the most similar candidates. For products in the top percentile of similarity, the probability of addition exceeds 4 percent per year, substantially higher than the average effect implied by the linear specification. In contrast, for products below approximately the 80th percentile, the probability of addition is not statistically different from zero.

This profile is the empirical counterpart of the cost-curvature parameter η in the adapted framework of Section 2. The adding probability takes the form $P(\text{Add}_{j\omega} = 1 \mid \Phi) = 1 - G(\kappa W(1 - \Phi)^\eta)$ in Equation (6). The linear specification in Equation (15) estimates the average slope of this relationship, which the model predicts to be positive (Equation (7)); the percentile profile instead traces its shape. A steep near-core cost gradient; formally $\eta < 1$, renders $P(\text{Add}_{j\omega} = 1 \mid \Phi)$ convex in Φ , concentrating viable additions among the candidates most similar to the core. The observed pattern, near-zero addition probability below roughly the 80th percentile followed by a sharp spike at the top percentile, is precisely this convexity: it identifies pronounced curvature ($\eta < 1$). The gap between the 1.3 percentage point average

linear effect and the top-percentile effect of roughly 4 percent suggests that the mass of additions sits at the very top of the similarity distribution.

The threshold and the curvature are distinct objects in the model, and the figure separates them. That additions occur only above a certain similarity reflects the stochastic threshold $\Phi_{j\omega}^*$ generated by demand and cost heterogeneity across firm–product pairs, whereas the concentration among the very most similar candidates reflects the cost curvature $\eta < 1$. The percentile figure therefore does more than support a threshold interpretation: it pins down the shape of the near-core cost function that the framework builds on.

I next examine product dropping behavior further. For incumbent products, similarity levels are compressed and relatively high, so it is rank rather than the similarity level that carries the economically relevant within-portfolio variation; I therefore group incumbents by rank rather than by similarity bin. Hence, I estimate:

$$P(\text{Drop}_{j\omega,t+1}) = \sum_{i=2}^{|\Omega_{jt}|} \beta_i \mathbf{1}\{\text{rank}_{j\omega t} = i\} + \lambda_j + \lambda_t + \lambda_{\bar{\omega}_j} + \varepsilon_{j\omega,t+1}, \quad (18)$$

where the first ranked (core) product is the omitted reference, and candidate-product fixed effects are omitted since rank replaces the similarity regressor for incumbents. Figure 7 plots the probability of dropping a product as a function of its rank within the firm’s product portfolio.

The figure shows a clear increasing relationship: lower-ranked products are significantly more likely to be dropped. The second-ranked product in firm-year sales has an almost-zero probability of being dropped. However, this increases to nearly 5 percent for the third-ranked product. Since lower-ranked products are also less similar to the firm’s core product (as shown in Figure 5), this provides additional evidence that firms exit products that are both less important and more distant in task space. Because the upper limit of the sum, $|\Omega_{jt}|$, varies across firms, the high-rank coefficients are identified from progressively fewer, wider-scope firms and are correspondingly less precisely estimated.

Taken together, these findings suggest that product scope adjustments are not random, but follow a systematic pattern. Firms expand along paths of high task similarity and contract along paths of low similarity. This behavior is consistent with a model in which firms

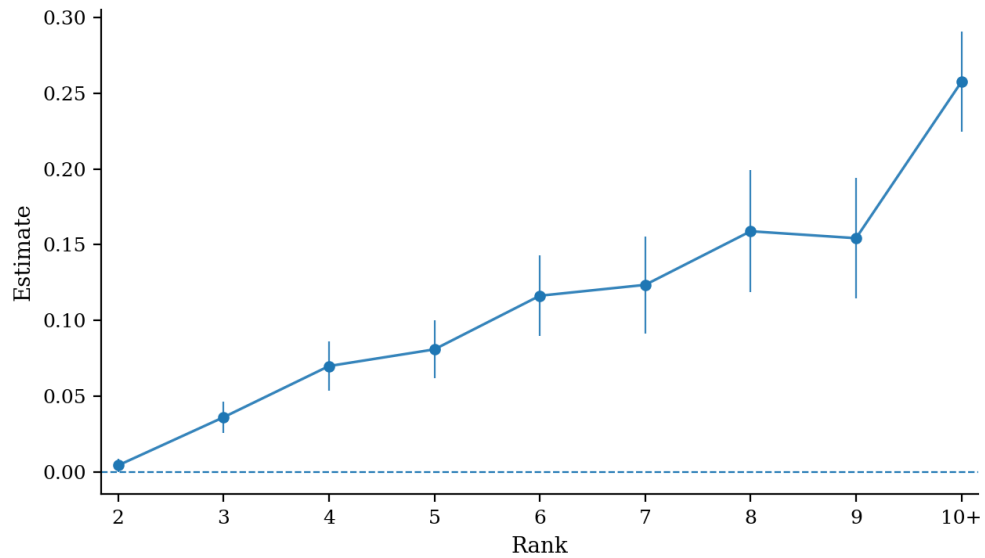


Figure 7: *This figure displays the probability of product dropping as a function of product rank within the firm.*

face increasing costs of adapting to distant products and therefore optimally concentrate production around their core competencies.

7 Conclusion

This paper studies how MPFs organize and adjust their product scope through the lens of occupational similarity. The central idea is that products differ in the tasks required for production, and that firms face lower costs when expanding into products that draw on capabilities already embedded in their workforce. By linking detailed employer–employee data to firm-level product information, the paper develops a task-based measure of product relatedness and uses it to analyze product scope decisions.

Methodologically, the paper contributes by constructing product-level vectors of occupational requirements using SPFs. These vectors provide a data-driven representation of the task content of production at the product level. Building on this, I introduce a similarity measure that extends the standard cosine similarity approach by incorporating a weighting matrix based on bilateral occupational skill transferability. This weighting matrix, estimated from excess worker flows, allows occupations to be partially related rather than strictly identical. For instance, metal workers and welders share high estimated skill transferability with blacksmiths and toolmakers. As a result, the similarity measure captures not only direct overlap in occupational inputs, but also more subtle forms of task similarity across products.

The resulting product space exhibits economically meaningful structure. Products cluster into groups that reflect shared production requirements, while also displaying strong cross-category linkages that are not captured by standard product classifications. At the same time, the overall distribution of similarities is sparse, indicating that most products share limited overlap in task content. These patterns suggest that the proposed measure proxies for underlying production technologies in a way that is both flexible and informative.

The empirical analysis of MPFs shows that occupational similarity is a key determinant of product scope decisions. Firms systematically concentrate production in products that are close to their core competencies in task space. The adding and dropping margins, however, operate asymmetrically. Product additions are strongly directed toward highly similar products and are concentrated in the upper percentiles of the similarity distribution, with addition probabilities indistinguishable from zero for less similar candidates. Product dropping, by contrast, is governed primarily by within-portfolio rank: products that are

more distant from the core are more likely to be dropped, particularly when they account for a small share of firm output. These findings are consistent with the adapted framework of Section 2, in which firms expand along dimensions where adaptation costs are low and contract where these costs are high. The concentration of additions in the very upper tail, beyond the positive average effect, reflects the curvature of the near-core cost function ($\eta < 1$) rather than a threshold alone.

Taken together, the results highlight a clear mechanism of firm growth: MPFs expand by reusing and recombining existing capabilities rather than by continuously developing entirely new ones. This provides a task-based foundation for understanding diversification and supports the view that the reusability of existing capabilities shapes firm scope. More broadly, the approach developed in this paper offers a framework for studying production structure using worker-level data, and can be applied to a range of questions related to firm dynamics, technological change, and the organization of production.

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Appendix A Theoretical Framework Appendices

A.1 Similarity threshold for production

I start from Equation 2, and consider the case where the inequality is binding:

$$a' - W \cdot [\gamma_\omega^0 + \kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta] - b'e[X_j + Y] = 0$$

Then, I add the cost function to both sides:

$$W \cdot [\gamma_\omega^0 + \kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta] = a' - b'e[X_j + Y]$$

And divide by W :

$$\gamma_\omega^0 + \kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta = \frac{a' - b'e[X_j + Y]}{W}$$

And subtract γ_ω^0 :

$$\kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta = \frac{a' - b'e[X_j + Y]}{W} - \gamma_\omega^0$$

Next, I divide by κ , and multiply W to γ_ω^0 on RHS:

$$(1 - \Phi(\bar{\omega}_j, \omega))^\eta = \frac{a' - b'e[X_j + Y] - W\gamma_\omega^0}{\kappa W}$$

And take the $1/\eta$ 'th root:

$$1 - \Phi(\bar{\omega}_j, \omega) = \left(\frac{a' - b'e[X_j + Y] - W\gamma_\omega^0}{\kappa W} \right)^{\frac{1}{\eta}}$$

Lastly, I subtract 1 on both sides, multiply by -1 , and denote the threshold as $\Phi^*(\cdot)$ holding all terms constant:

$$\Phi^*(\bar{\omega}_j, \omega) = 1 - \left(\frac{a' - b'e[X_j + Y] - W\gamma_\omega^0}{\kappa W} \right)^{\frac{1}{\eta}}$$

With the following properties:

$$\frac{\partial \Phi^*(\cdot)}{\partial a'} < 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial b'} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial X} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial Y} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial W} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial \kappa} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial \gamma_\omega^0} > 0$$

Appendix B Sample Summary Appendices

B.1 Firms and product summary statistics

This appendix presents some summary statistics related to firms and products in the sample (see Section 3.1.1. An executive overview is provided below in Table 6.

Table 6: Firm-level summary statistics

Firms	Mean revenue	Mean value added	Mean employees	Mean years worked	Mean products (HS2)	Mean products (HS4)	Mean products (HS6)
3,917	158.5M	51.4M	72.8	66.5	1.61	2.28	2.91

Notes: The table reports average firm characteristics across the sample, including revenue, value added, employment, years worked, and product scope across HS aggregation levels.

The mean number of products, however, masks a right skewed distribution: Table 7 displays the distribution of firms in the sample across discrete product mix size categories.

Table 7: Distribution of firms by number of products

Number of products	Firms	Share
1	2,090	0.533
2	839	0.214
3	350	0.089
4	178	0.045
5	108	0.028
6	71	0.018
7	48	0.012
8	32	0.008
9	21	0.005
≥ 10	185	0.047

Notes: This table contains the number of firms producing between 1 and more than 10 products in discrete categories. Shares are proportions of total firms. Product counts are computed across the full sample period from 2010 to 2019.

Of course, products are not produced equally frequent or equally intensively. Table 8 contains the top 10 products sorted by two margins: 1) number of manufacturers, and 2)

mean sales.

Table 8: Top HS4 products by number of manufacturers and by mean sales

Ranked by number of manufacturers				Ranked by mean sales			
Rank	p	N_p	μ_p^{sales}	Rank	p	N_p	μ_p^{sales}
1	7308: steel structures	368	11,058	1	7308: steel structures	368	11,058
2	7326: iron/steel articles	340	6,205	2	2309: animal feed	47	8,393
3	3926: plastic articles	214	4,062	3	9403: furniture	207	8,130
4	7204: scrap iron/steel	214	152	4	9021: orthopedic devs	35	6,713
5	9403: furniture	207	8,130	5	7326: iron/steel articles	340	6,205
6	8479: machinery n.e.s.	205	3,830	6	4418: wood joinery	109	5,656
7	4911: printed matter	129	2,113	7	6810: cement articles	68	5,205
8	3923: plastic packaging	117	3,726	8	1905: bakery products	45	5,063
9	8428: lifting machinery	115	3,036	9	9031: measuring instr.	48	4,835
10	4418: wood joinery	109	5,656	10	8481: valves	60	4,827

Notes: This table reports the top 10 product categories (4 digits), denoted p , under two alternative rankings: by total number of manufacturers, denoted N_p , and by mean sales, denoted μ_p^{sales} . The latter is reported in thousands DKK in 2015 values.

Product do not appear equally frequent in the same product mix, either. Table 9 displays the top 10 most frequently co-produced product pairs in the sample.

Table 9: Most common co-production pairs

Rank	ω_1	ω_2	N_{ω_1, ω_2}	$N_{\omega_1, \omega_2}/N$
1	7204: Iron and steel scrap	7308: Steel structures	81	0.0207
2	7204: Iron and steel scrap	7326: Steel articles	78	0.0199
3	7308: Steel structures	7326: Steel articles	64	0.0163
4	4901: Printed books	4911: Printed materials	43	0.0110
5	7326: Steel articles	7616: Aluminium articles	43	0.0110
6	9401: Seats	9403: Furniture	43	0.0110
7	3926: Plastic articles	7326: Steel articles	42	0.0107
8	8428: Lifting equipment	8431: Machinery parts	37	0.0094
9	4902: Newspapers	4911: Printed materials	34	0.0087
10	3926: Plastic articles	8480: Moulds and dies	31	0.0079

Notes: This table reports the most common product pairings. N_{ω_1, ω_2} denotes the number of firms producing both products at overlapping times during the sample period. N denotes the total number of unique firms present in the sample. Shares are proportions of all firms.

B.2 Worker and occupations summary statistics

This appendix presents some summary statistics related to workers and occupations in the sample (see Section 3.1.2).

Table 10 contains brief summary statistics of sample workers.

Table 10: Worker-level summary statistics

	Gender	Workers	Years	Wage	Hours	Hourly wage	Occs (3-digit)
Mean	1.29	370,072	4.42	358,338.17	1,412.09	372.59	1.25

Notes: The table reports worker-level means. Wage is reported in levels, and hourly wage is computed as wage divided by hours. Occupation measures refer to the number of distinct occupations at the 3-digit level.

Most firms employ workers across roughly 10 occupations at the three-digit level. Figure 13 displays the kernel density of organizational complexity, i.e. number of unique occupations, for all firms in the sample.

Clearly, specific occupations are not equally present in firms either. Table 11 displays the top 10 most frequently present occupations among sample firms.

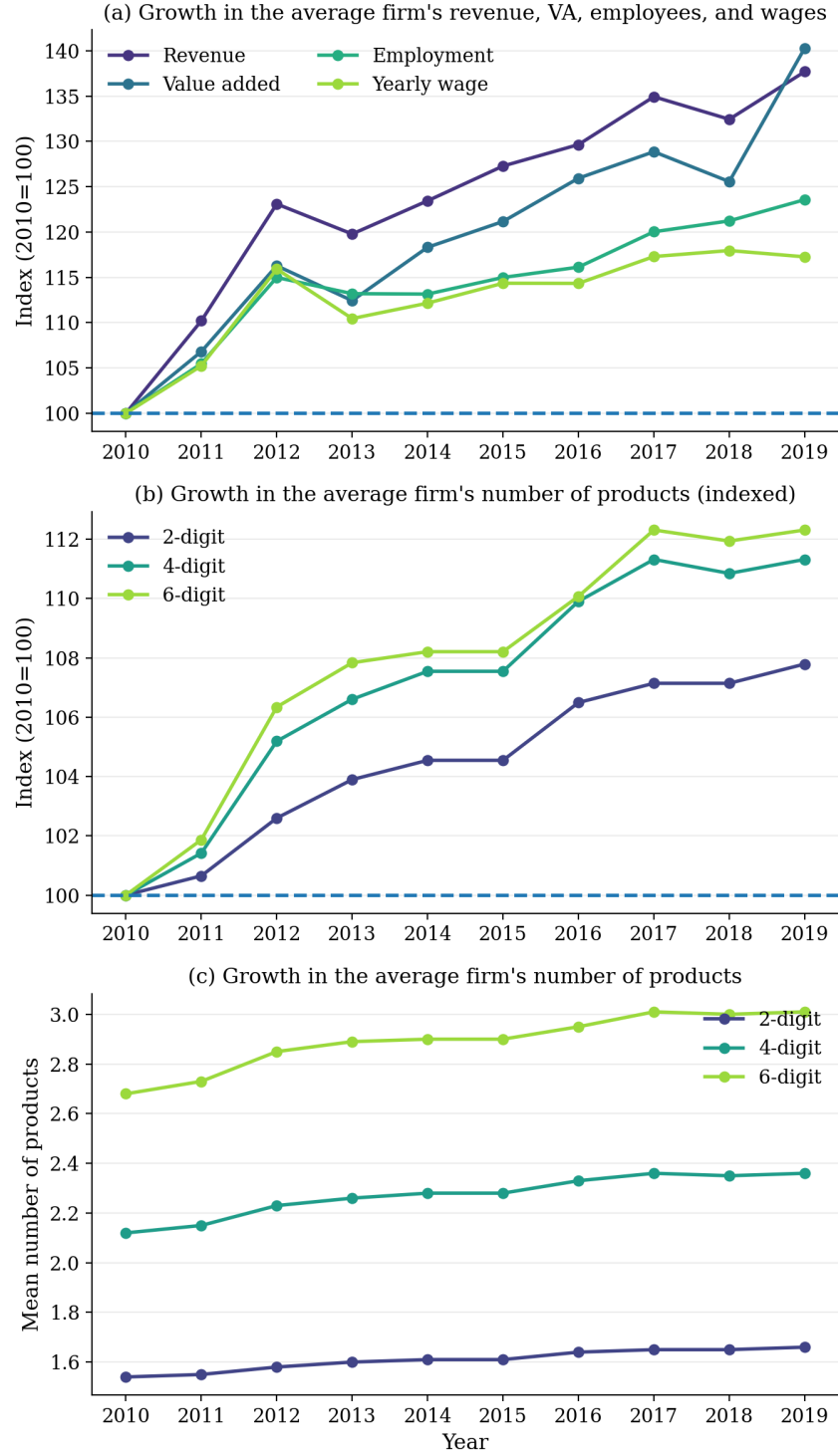


Figure 8: This figure shows (a) the growth of revenue, value added (VA), employees, and yearly wages for the average firm (2010 = 100), (b) the growth in the average number of products manufactured (2010 = 100) across different aggregation levels of products, and (c) the numeric growth of panel (b).

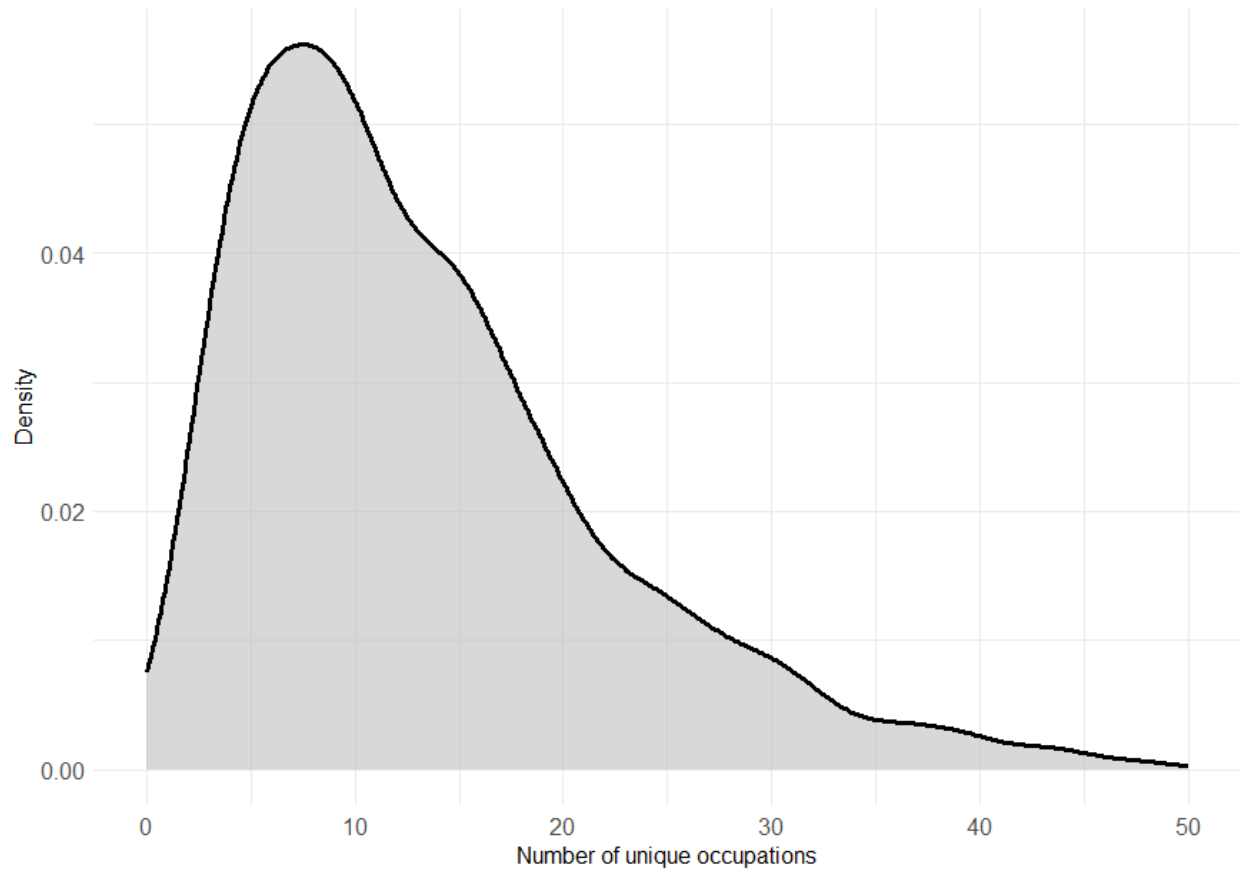


Figure 9:

B.3 Further statistics for stylized facts

Table 11: Most prevalent occupations

DISCO	<i>N</i> firms	Share
112 Managing Directors & Chief Executives	2,684	0.757
132 Manufacturing, Construction & Distribution Managers	2,271	0.641
411 General Office Clerks	2,093	0.591
311 Physical & Engineering Technicians	2,092	0.590
332 Sales & Purchasing Agents/Brokers	1,833	0.517
331 Financial & Mathematical Associates	1,812	0.511
722 Blacksmiths & Toolmakers	1,778	0.502
122 Sales, Marketing & Development Managers	1,646	0.464
121 Business Services & Admin Managers	1,641	0.463
432 Materials & Transport Clerks	1,495	0.422

Notes:

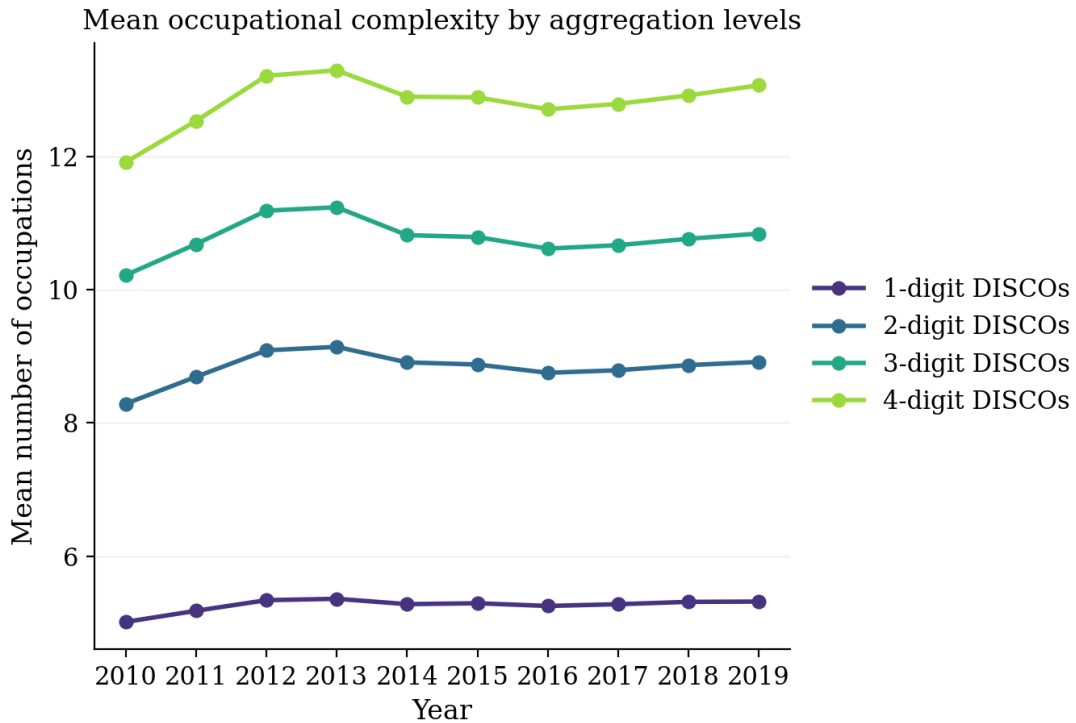


Figure 10:

Appendix C SPF Definitions

C.1 Single and multi product firms

The strictest definition of an SPF is a firm that produces exclusively within a single product category, such that 100 percent of its sales are concentrated in that category. In practice,

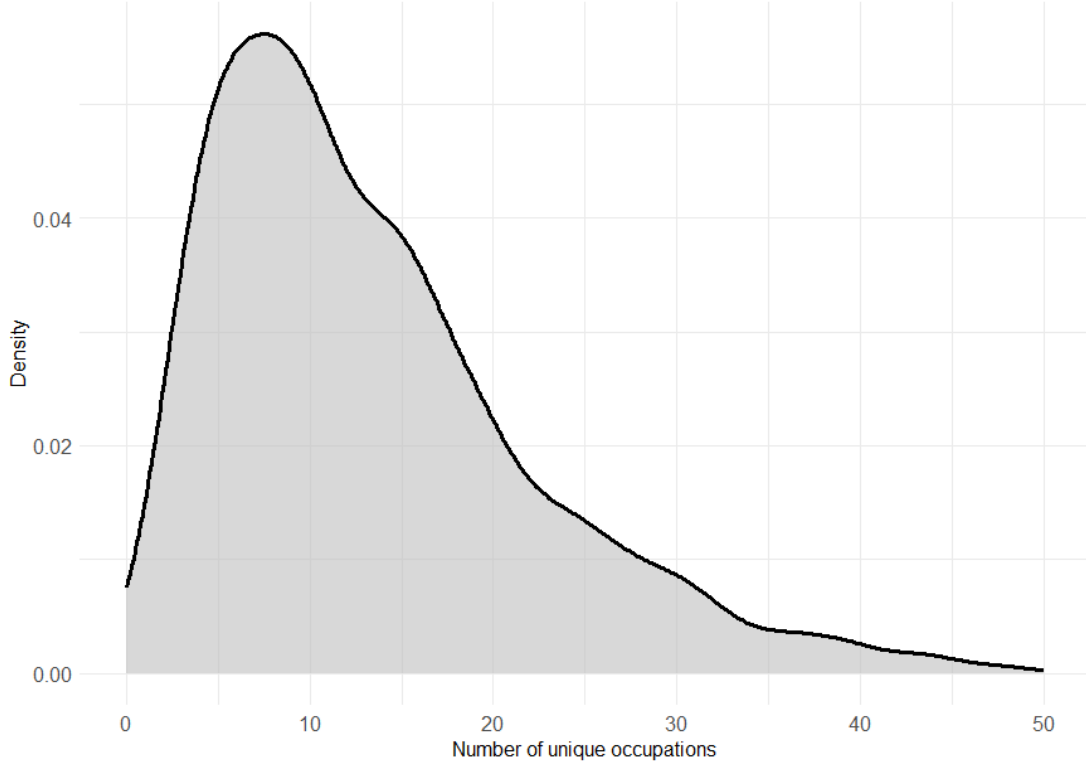


Figure 11:

however, many firms generate at least a small fraction of sales outside their core product. As a result, the strict SPF definition covers only a limited and products, leaving much of the product space uncovered. A tradeoff therefore arises: stricter definitions yield cleaner occupation–product vectors but sacrifice coverage, while looser definitions increase coverage at the cost of contamination from co-production. In Appendices C.2, I present an alternative approach to relaxing the strict SPF definition.

To fix notation, define the sales share of product ω in firm j ’s portfolio as:

$$s_{j\omega} = \frac{X_{j\omega}}{X_j}. \quad (19)$$

This measure captures the relative importance of product ω for firm j . For ease of notation, I omit the time index t , although these shares are observed annually.

The canonical SPF definition requires concentration in a single product category. For-

mally:

$$\text{SPF}_j = \begin{cases} 1 & \text{if } \max_{\omega \in \Omega_j} s_{j\omega} \geq \alpha, \\ 0 & \text{otherwise,} \end{cases} \quad (20)$$

where $\alpha \in (0, 1]$ is a threshold parameter. If $\text{SPF}_j = 0$, then $\text{MPF}_j = 1$ by construction, so every firm is classified as either SPF or MPF. Under the strict definition ($\alpha = 1$), only a small share of products are produced by SPFs. A natural relaxation is to lower α , which increases both the share of firms classified as SPFs and the coverage of products. This highlights the central tradeoff: stricter definitions reduce contamination but limit coverage, whereas looser definitions improve coverage but introduce noise from co-production.¹² Figure 13 illustrates product coverage as a function of $\alpha \in (0.5, 1]$. Lower values of α expand SPF coverage of products (panel (b)), but at the expense of less pure occupation–product vectors, as co-produced activities receive greater weight (panel (d)). Panel (c) shows the share of total sales accounted for by SPF-produced products. Even under the strictest definition, these products account for a large share of economic activity, with nearly 90 percent of all sales coming from approximately 40 percent of products.

A key advantage of using SPF occupational vectors is that they rely on observed production data. The identifying assumption is that the observed occupational distribution reflects the tasks required to produce the firm’s core product. Firms that produce exclusively within a single product category therefore provide the cleanest mapping from occupations to product-level task requirements. However, as shown in Figure 13, approximately 60 percent of firms produce entirely within one product code, yet these firms cover only 39 percent of products. While these products account for a large share of total economic activity (89 percent), these firms are relatively small and represent less than 35 percent of total sales (see Appendix C.1.1, panel (d)).

Even under the strict definition, there remains a case for relaxing the SPF criterion. The sharp decline in the share of SPFs when $\alpha > 0.95$ suggests that many firms produce a small fraction of output outside their core product. I argue that such firms are not representative of the canonical notion of multiproduct firms in the literature, but rather reflect minor

¹²The flipside of having more SPFs is fewer MPFs. Hence, if there is a subsequent aim to analyze MPFs, this tradeoff may also affect sample size in later stages.

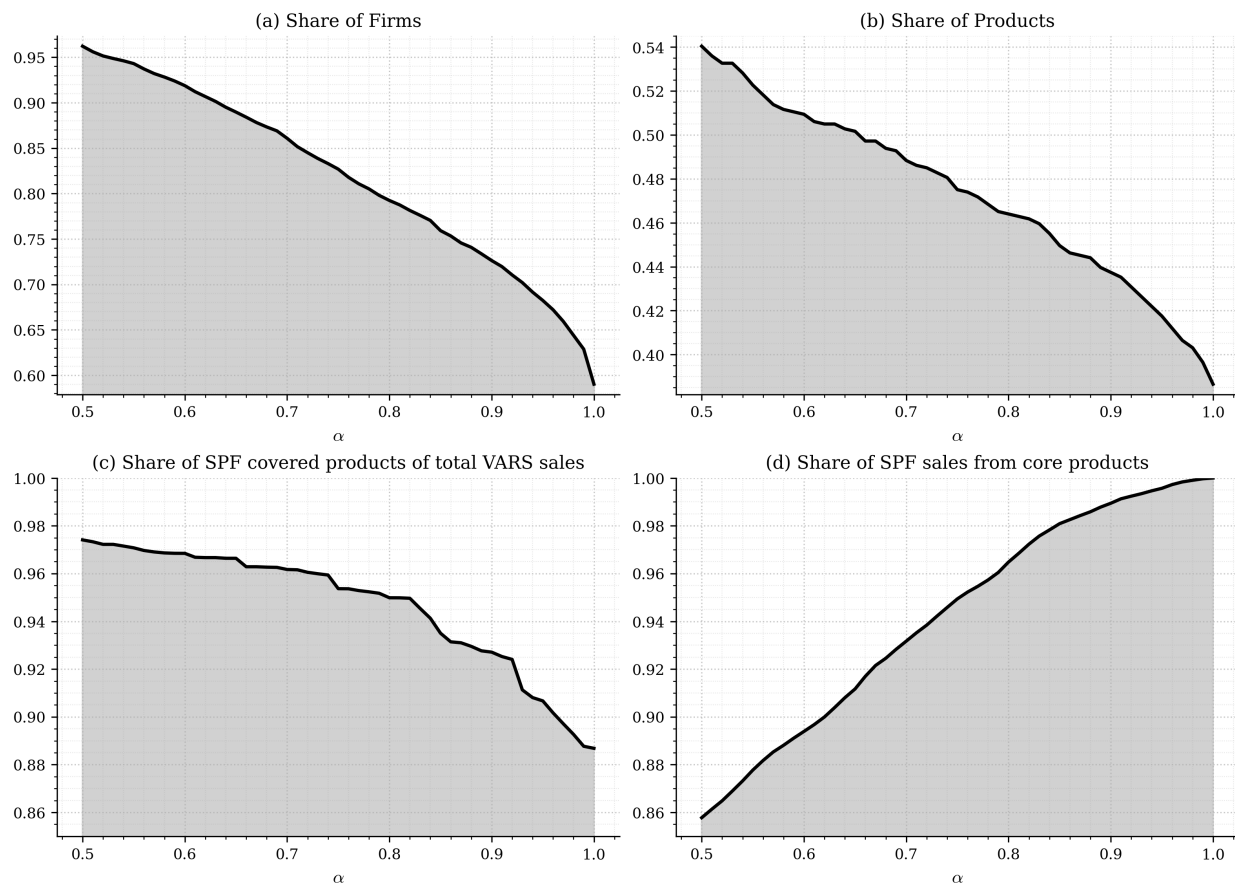


Figure 12:

deviations from otherwise concentrated production structures.

Appendix C.2 introduces an alternative approach to SPF definitions, where core-product thresholds are evaluated at different aggregation levels than those used to define the core product itself. For example, a firm may be classified as an SPF if it produces 100 percent within a single two-digit product category, while its core product is assigned at the four-digit level.

In practice, the preferred SPF definition balances coverage and purity: it should maximize product coverage while ensuring that occupational vectors remain representative of production tasks. I therefore adopt a definition where firms are classified as SPFs if at least 90 percent of their sales are concentrated in a single product category. Under this definition, SPFs account for 73 percent of firms and cover 44 percent of the product space through their core products. These products account for more than 92 percent of total sales, imply-

ing substantial coverage. Moreover, core products account for more than 98 percent of SPF sales, indicating that the resulting occupational vectors remain sufficiently clean.

C.1.1 Additional SPF metrics

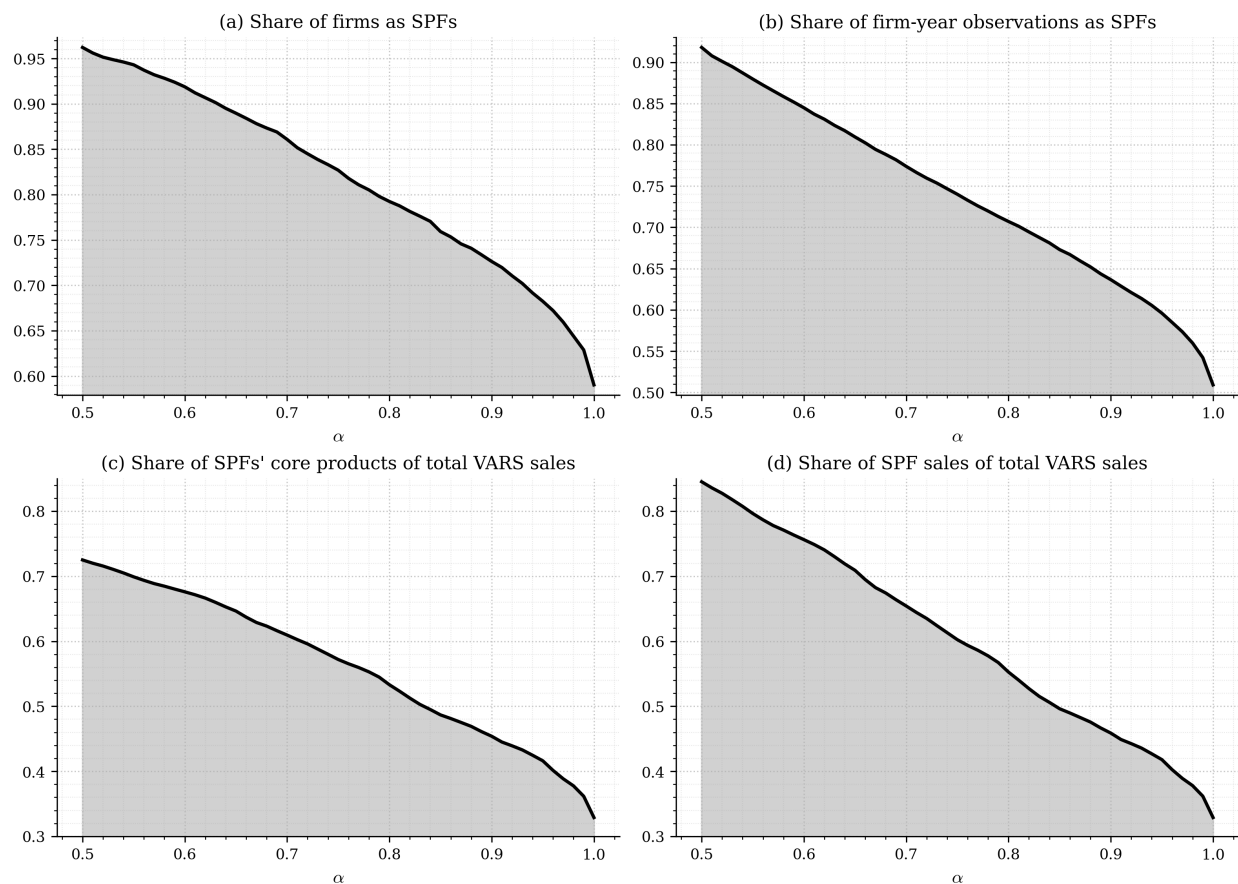


Figure 13:

C.2 Alternative SPF definition approach

A second approach to broadening the SPF concept is to classify firms at a coarser product aggregation level, which is one way to accepting a little co-production while ensuring cleanliness. In this case, a firm is considered an SPF if most of its sales fall within a single product group at aggregation level k , even if these sales are spread across several sub-products within that group. Hence, SPF classifications are constructed at one level of HS aggregation following Equation 20, while assignment of SPF core product is carried out at another level, denoted p . For instance, k in Equation 20 may refer to a 2-digit grouping, in which case all sales within a single two-digit code are aggregated to test the SPF condition. Assignment of a firm's core product is then determined by the largest constituent product within the chosen group k^* . The product at level p which has the highest sales share is then defined as the firm core-product, for which further analysis is conducted:

$$\text{If } \text{SPF}_f^k = 1 \quad \Rightarrow \quad \text{SPF}_{fp} = \begin{cases} 1 & \text{if } p \in k^* \text{ and } \theta_{fp} = \max_{p \in k^*} \theta_{fp}, \\ 0 & \text{otherwise,} \end{cases} \quad (21)$$

where $k^* \in HS^{(k)}$ is the core group for firm f , and $\theta_{fp}^m = s_{fp}^m / S_f^m$ are sales shares defined level m .

Most obviously, one would specify p to be at a finer level than k : $p > k$. One additional advantage of this method, is that the relaxation at level k should not lead to excessive contamination at level m due to co-production of very distant products. As an example, consider a firm producing only within a two-digit code "grain products" (code 10), which will be classified as a SPF. Then, if the majority of sales is coming from one four-digit code "rice" (code 1006), this method will assign this firm as a SPF of product code 1006.

Table 12 displays the coverage of all possible levels of k and p in Equation 21, where $(k, p) \in \{2, 4, 6\} \times \{2, 4, 6\}$ are digit lengths. I report the share of firms defined as SPFs at this level, the products covered at aggregate k level and detailed p level, and the share of sales coming from SPFs. Importantly, the table also includes the share of all manufacturing sales from the products covered by SPF core products, and the share of SPF sales coming from

Table 12: Two-level SPF definition coverage

Aggr. (k^*)	Core (p)	Firms (share)	Products (aggr.)	Products (det.)	Sales (share)	Coverage p (of all sales)	Cleanness p (of SPF sales)
HS2	HS2	65.5%	77.1%	77.1%	41.9%	99.5%	100.0%
HS2	HS4	65.5%	77.1%	35.7%	41.9%	91.5%	93.9%
HS2	HS6	65.5%	77.1%	23.7%	41.9%	73.6%	89.4%
HS4	HS2	53.0%	31.4%	77.1%	29.4%	99.5%	100.0%
HS4	HS4	53.0%	31.4%	31.4%	29.4%	88.4%	100.0%
HS4	HS6	53.0%	31.4%	20.2%	29.4%	70.7%	94.7%
HS6	HS2	45.3%	17.7%	75.9%	21.8%	99.2%	100.0%
HS6	HS4	45.3%	17.7%	29.2%	21.8%	86.1%	100.0%
HS6	HS6	45.3%	17.7%	17.7%	21.8%	65.0%	100.0%

Notes: Percentages are rounded to one decimal place. “SPF” denotes single-product firms. Column k^* refers to HS chapters at the 2-, 4-, or 6-digit level; and column p is the HS level used to classify a firm’s core product following Equation 21. Column three, “Firms (share)” denotes the share of total firms defined as SPFs, column 4 “Products (aggr.)” displays the number of products at level k covered by SPFs, column 5 “Products (det.)” the share of products covered by SPFs at level p , and column 6 “Sales (share)” the share of total sales registered by SPFs. The last two columns respectively show the coverage of SPF products; i.e. the share of total sales by all manufacturing firms that is covered by SPF products, and the cleanness of SPF sales; i.e. the share of SPF sales registered by their core-product.

core products. The former attempts to address the economic importance of the products covered by SPFs, while the latter addresses the cleanness of occupational vectors, as more clean measures of occupations needed to produce a product implies little production outside of the core product.

Appendix D Bilateral Product Similarities and Occupational Flows

D.1 A numeric example of unweighted and weighted cosine similarities

This appendix contains calculations based on the example presented and illustrated in Section 4 in Figure 3. The numerical illustration is:

$$\mathcal{O} = \begin{pmatrix} 0.06 & 0.06 & 0.10 & 0.18 & 0.10 \\ 0.12 & 0.04 & 0.02 & 0.04 & 0.16 \\ 0.04 & 0.18 & 0.14 & 0.04 & 0.00 \end{pmatrix}$$

, where rows represent products, and columns represent occupations.

The corresponding cosine similarities would hence be:

$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle = 0.6827$$

$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_3) \rangle = 0.5997$$

$$\langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_3) \rangle = 0.3343$$

such that $\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle > \langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_3) \rangle > \langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_3) \rangle$.

Then, consider a weighting matrix Ψ where rows correspond to origin-occupations o and columns to destination-occupations o^* , which captures the estimated skill-transferability

from an origin occupation o to a destination occupation o^* :

$$\Psi = \begin{pmatrix} 1 & 0.1 & 0 & 0 & 0 \\ 0.1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0.5 & 0 \\ 0 & 0 & 0.5 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

, where both rows and columns are occupations. Occupations 1 and 2 share skills or tasks to a small degree, and occupations 3 and 4 share skills to a larger degree. Occupation 5 does not share skills with any other occupation. Now, occupations are not only compared piecewise, but I compare all occupations across. Hence, for two products, all possible bilateral occupation pairs can increase the similarity, if there is presence and skill-sharing. The updated weighted similarities are now:

$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle = 0.6637$$

$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_3) \rangle = 0.7207$$

$$\langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_3) \rangle = 0.4125$$

, and hence the new ranking of similarities is $\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_3) \rangle > \langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle > \langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_3) \rangle$.

This example hence serves to illustrate the importance of considering substitutability or skill-sharing between detailed occupation pairs, as products ω_1 and ω_3 now share more occupational overlap than ω_1 and ω_2 . In general, the occupational similarity magnitude is hence determined by the relative importance and the size of the estimated skill-sharing.

To formally write out the calculations, consider first the unweighted cosine similarity between ω_1 and ω_2 :

$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle = \frac{\mathcal{O}(\omega_1)^\top \mathcal{O}(\omega_2)}{\|\mathcal{O}(\omega_1)\| \cdot \|\mathcal{O}(\omega_2)\|}.$$

The inner product is:

$$\begin{aligned}
\mathcal{O}(\omega_1)^\top \mathcal{O}(\omega_2) &= 0.06 \cdot 0.12 + 0.06 \cdot 0.04 + 0.10 \cdot 0.02 + 0.18 \cdot 0.04 + 0.10 \cdot 0.16 \\
&= 0.0072 + 0.0024 + 0.0020 + 0.0072 + 0.0160 \\
&= 0.0348.
\end{aligned}$$

The magnitudes are:

$$\begin{aligned}
\|\mathcal{O}(\omega_1)\| &= \sqrt{0.06^2 + 0.06^2 + 0.10^2 + 0.18^2 + 0.10^2} = \sqrt{0.0596}, \\
\|\mathcal{O}(\omega_2)\| &= \sqrt{0.12^2 + 0.04^2 + 0.02^2 + 0.04^2 + 0.16^2} = \sqrt{0.0436}.
\end{aligned}$$

Thus,

$$\begin{aligned}
\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle &= \frac{0.0348}{\sqrt{0.0596} \sqrt{0.0436}} \\
&= 0.6827.
\end{aligned}$$

Next, consider the weighted cosine similarity:

$$\Phi(\omega_1, \omega_2) = \frac{\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle_\Psi}{\sqrt{\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_1) \rangle_\Psi} \sqrt{\langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_2) \rangle_\Psi}}$$

The numerator is:

$$\begin{aligned}
\mathcal{O}(\omega_1)^\top \Psi \mathcal{O}(\omega_2) &= 0.06(1 \cdot 0.12 + 0.1 \cdot 0.04) \\
&\quad + 0.06(0.1 \cdot 0.12 + 1 \cdot 0.04) \\
&\quad + 0.10(1 \cdot 0.02 + 0.5 \cdot 0.04) \\
&\quad + 0.18(0.5 \cdot 0.02 + 1 \cdot 0.04) \\
&\quad + 0.10(1 \cdot 0.16) \\
&= 0.0368.
\end{aligned}$$

The weighted magnitudes are:

$$\mathcal{O}(\omega_1)^\top \Psi \mathcal{O}(\omega_1) = 0.0796,$$

$$\mathcal{O}(\omega_2)^\top \Psi \mathcal{O}(\omega_2) = 0.0452.$$

Hence,

$$\begin{aligned} \langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle_\Psi &= \frac{0.0368}{\sqrt{0.0796} \sqrt{0.0452}} \\ &= 0.6637. \end{aligned}$$

Which is calculated from the following R-code:

```

1  # Function to calculate (unweighted) cosine similarity
2  fn_cosine_sim <- function(vec1, vec2) {
3
4      inner_prod <- sum(vec1 * vec2)
5      magnitude_v1 <- sqrt(sum(vec1^2))
6      magnitude_v2 <- sqrt(sum(vec2^2))
7
8      sim <- inner_prod / (magnitude_v1 * magnitude_v2)
9
10     return(sim)
11 }
12
13 # Function to calculate weighted cosine similarity
14 fn_weighted_cosine_sim <- function(vec1, vec2, mat_weights) {
15
16     # numerator
17     inner_prod <- t(vec1) %*% mat_weights %*% vec2
18
19     # denominator terms
20     magnitude_v1 <- sqrt(t(vec1) %*% mat_weights %*% vec1)
21     magnitude_v2 <- sqrt(t(vec2) %*% mat_weights %*% vec2)
22
23     # output

```

```

24     sim <- inner_prod / (magnitude_v1 * magnitude_v2)
25
26     return(as.numeric(sim))
27 }
28
29 # Example vectors: shares of labor input across occupations
30 v01 = c(0.06, 0.06, 0.10, 0.18, 0.10)
31 v02 = c(0.12, 0.04, 0.02, 0.04, 0.16)
32 v03 = c(0.04, 0.18, 0.14, 0.04, 0.00)
33
34 # Example weighting matrix (occupation similarity)
35 mWeights <- matrix(c(
36     1,    0.1, 0,    0,    0,
37     0.1, 1,    0,    0,    0,
38     0,    0,    1,    0.5, 0,
39     0,    0,    0.5, 1,    0,
40     0,    0,    0,    0,    1
41 ), nrow = 5, byrow = TRUE)
42
43 # --- Unweighted cosine similarities ---
44 fn_cosine_sim(v01, v02) # 0.6827
45 fn_cosine_sim(v01, v03) # 0.5997
46 fn_cosine_sim(v02, v03) # 0.3343
47
48 # --- Weighted cosine similarities ---
49 fn_weighted_cosine_sim(v01, v02, mWeights) # 0.6637
50 fn_weighted_cosine_sim(v01, v03, mWeights) # 0.7207
51 fn_weighted_cosine_sim(v02, v03, mWeights) # 0.4125

```

D.2 Flow counts per year

Table 13: Job switchers by year

Year	Workers	Switchers	Share
2011	1,490,079	141,024	0.0946
2012	1,489,660	115,478	0.0775
2013	1,491,990	117,488	0.0787
2014	1,491,852	153,843	0.1031
2015	1,514,741	127,323	0.0841
2016	1,496,945	124,446	0.0831
2017	1,479,878	123,971	0.0838
2018	1,488,514	121,239	0.0814
2019	1,522,072	122,917	0.0808

Notes: Number of workers, number of switching to a new occupation, and the resulting switch share by year.

D.3 Sankey plots of occupational flows

D.4 Flow counts within or between employers

Table 14: Job switches by hierarchical direction

	Up hierarchy		Total
	No	Yes	
Employer switch: No	4,637	4,290	8,927
Employer switch: Yes	7,014	6,979	13,993
Total	11,651	11,269	22,920

Notes: This table reports the number of worker flows between occupations, split into flows within and between employers, and whether the flow is up or down the numerical hierarchy of the occupation classifications (see <https://www.dst.dk/da/Statistik/dokumentation/nomenklaturer/discoloen>).

D.5 Histogram of bilateral occupation pairs flow count

Flows to and from occupation 112

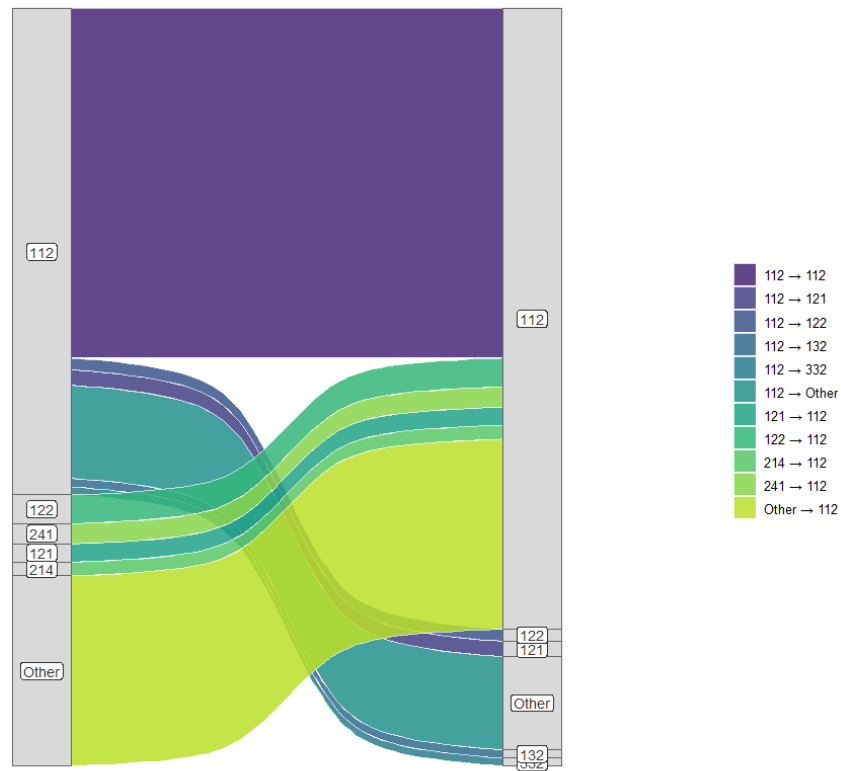


Figure 14: *This figure displays the in- and outflows of occupation 112: Admin managers.*

Flows to and from occupation 214

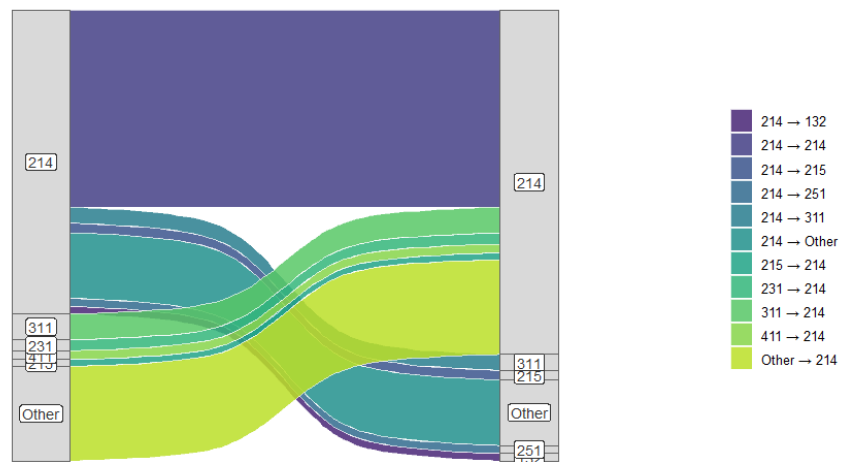


Figure 15: *This figure displays the in- and outflows of occupation 214: Engineers.*

Flows to and from occupation 411

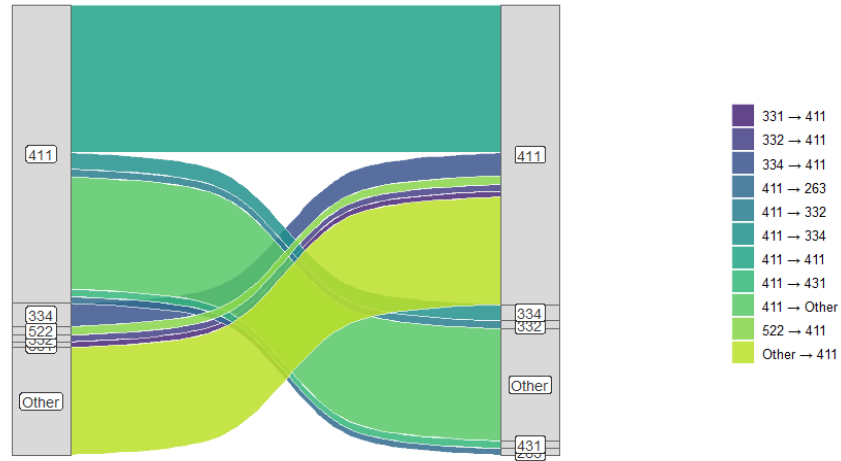


Figure 16: *This figure displays the in- and outflows of occupation 411: Office clerks.*

Flows to and from occupation 821

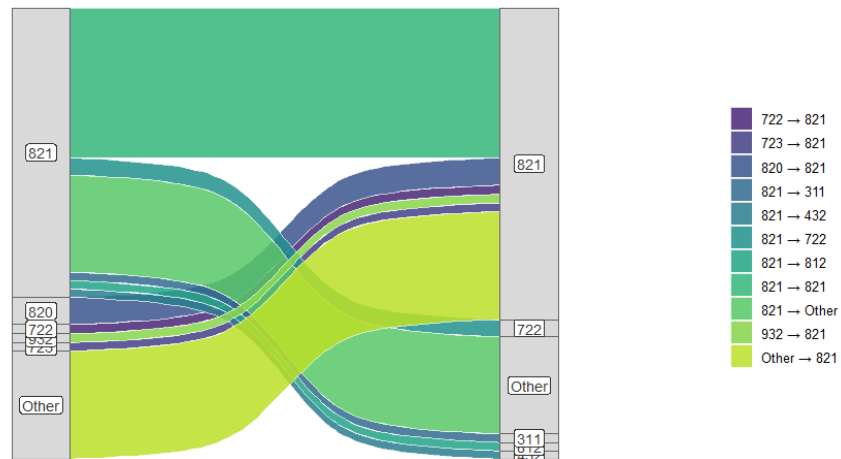


Figure 17: *This figure displays the in- and outflows of occupation 821: Assembly work.*

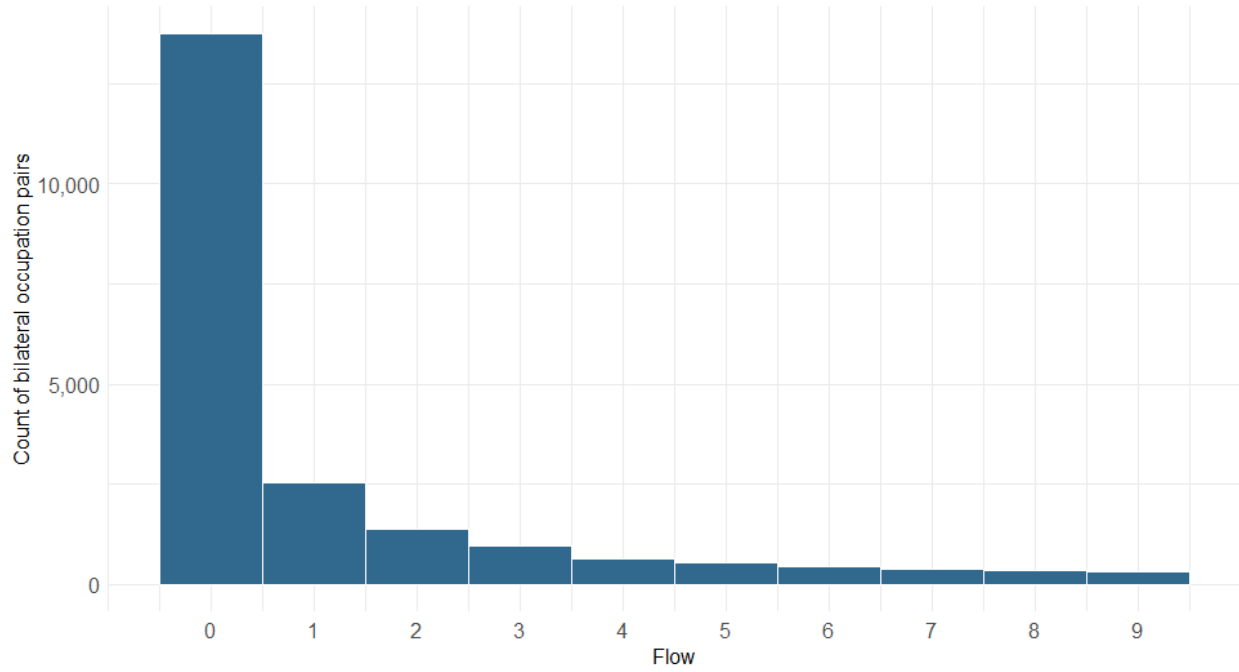


Figure 18:

D.6 Estimation of bilateral occupation skill-transfer

	Coefficient
<i>Count model</i>	
Intercept	0.94*** (0.02)
total employment _{<i>i</i>}	1.05*** (0.03)
total employment _{<i>i</i>*}	0.91*** (0.02)
avg. wage _{<i>i</i>}	0.06** (0.02)
avg. wage _{<i>i</i>*}	0.16*** (0.02)
employment growth _{<i>i</i>}	−0.07*** (0.02)
employment growth _{<i>i</i>*}	0.09*** (0.03)
wage growth _{<i>i</i>}	−0.22*** (0.04)
wage growth _{<i>i</i>*}	0.25** (0.09)
Log(theta)	−1.53*** (0.02)
<i>Zero model</i>	
Intercept	−20.63*** (1.96)
total employment _{<i>i</i>}	−0.65*** (0.05)
total employment _{<i>i</i>*}	−37.44*** (3.31)
avg. wage _{<i>i</i>}	−0.15** (0.05)
avg. wage _{<i>i</i>*}	−0.39*** (0.06)
employment growth _{<i>i</i>}	−0.06 (0.07)
employment growth _{<i>i</i>*}	−0.18*** (0.05)
wage growth _{<i>i</i>}	−0.02 (0.12)
wage growth _{<i>i</i>*}	0.21*** (0.06)
AIC	66755.11
Log Likelihood	−33358.56
Number of Observations	23, 244
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$	

D.7 Top 30 occupationally similar products

Table 15: Most closely related HS product pairs

Rank	Product 1	Product 2	Similarity
1	7325: Cast iron articles	7419: Other copper articles	0.988
2	8207: Interchangeable tools	8480: Moulding boxes	0.986
3	8466: Machine tool parts	8480: Moulding boxes	0.981
4	1601: Sausages	1604: Prepared fish	0.980
5	7304: Steel pipes (seamless)	8466: Machine tool parts	0.979
6	8207: Interchangeable tools	8466: Machine tool parts	0.977
7	8431: Machinery parts	8437: Grain processing machinery	0.977
8	7326: Other iron/steel articles	8707: Vehicle bodies	0.976
9	7204: Ferrous scrap	7304: Steel pipes (seamless)	0.974
10	0201: Meat	1601: Sausages	0.973
11	8422: Dishwashing/packing machinery	8428: Lifting/handling equipment	0.973
12	8431: Machinery parts	8465: Woodworking machines	0.973
13	7204: Ferrous scrap	8707: Vehicle bodies	0.973
14	8465: Woodworking machines	8707: Vehicle bodies	0.972
15	7326: Other iron/steel articles	8437: Grain processing machinery	0.970
16	7326: Other iron/steel articles	8431: Machinery parts	0.970
17	7304: Steel pipes (seamless)	8480: Moulding boxes	0.970
18	8431: Machinery parts	8707: Vehicle bodies	0.968
19	0304: Fish fillets	1601: Sausages	0.967
20	3923: Plastic packaging	3926: Other plastic articles	0.966
21	1601: Sausages	1704: Sugar confectionery	0.965
22	4911: Other printed matter	8442: Printing machinery	0.964
23	3920: Plastic sheets	4819: Paper packaging	0.964
24	0201: Meat	1604: Prepared fish	0.963
25	7204: Ferrous scrap	8480: Moulding boxes	0.963
26	4821: Paper labels	4901: Printed books	0.963
27	7204: Ferrous scrap	8465: Woodworking machines	0.963
28	7204: Ferrous scrap	8207: Interchangeable tools	0.963
29	7309: Steel tanks	8609: Containers	0.963
30	7304: Steel pipes (seamless)	8207: Interchangeable tools	0.962

Notes: The table reports the most closely related HS4 product pairs based on a similarity measure. Higher values indicate greater similarity in production or input structure.

D.8 Distribution of bilateral product similarities

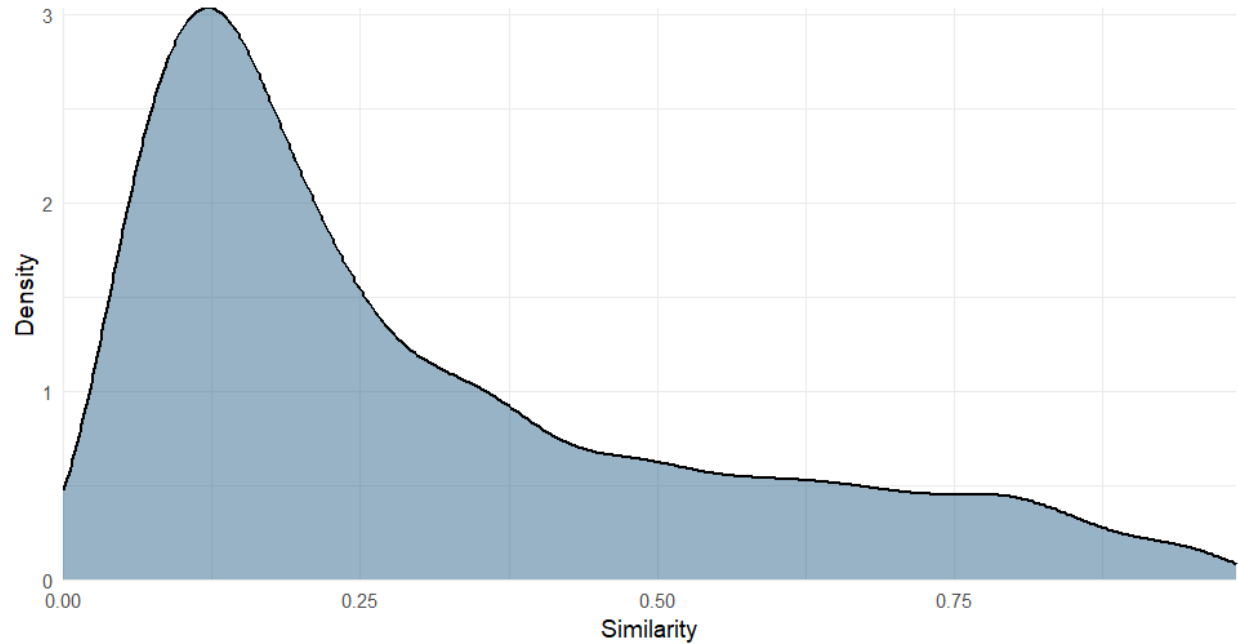


Figure 19: *This figure displays the distribution of similarities of bilateral product pairs across the GDPR approved product space also depicted in Figure 4 on the basis of 13. The distribution is clearly right-skewed and most frequently, product pairs have an occupational similarities around 0.2.*